

**A Project Report On**  
**“Ai Based Public Hazard Detection”**

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**In Fulfillment of the Requirement For**  
**Final Year of Bachelor of Technology**  
**In**

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**Hindi Seva Mandal's (Estd. 1950)**

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**COLLEGE OF ENGINEERING AND TECHNOLOGY**

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**Dr. Babasaheb Ambedkar Technological University Lonere**

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**CERTIFICATE**

This is to certify that **Mr. Farhan Zafar Deshmukh And Mr. Sushil Pandharinath Patil And Mr. Niraj Kishor Shirnath** has undergone and successfully completed their “**Project**” on “**Ai Based Public Hazard Detection**” for the fulfillment of **Final Year** of **Bachelor of Technology** in **Artificial Intelligence and Data Science** as prescribed by Dr. Babasaheb Ambedkar Technological University, Lonere during academic year **2025-2026**.

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# EXAMINERS CERTIFICATE

The Project Report Entitled: “**Ai Based Public Hazard Detection**” from Mr. Farhan Zafar Deshmukh, Mr. Sushil Pandharinath Patil and Mr. Niraj Kishor Shirnath,” In the fulfillment of **Final Year** for the award of degree of **Bachelor of Technology** in **Artificial Intelligence and Data Science** is certified.

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## **DECLARATION**

We hereby declare that the Project entitled, “**Ai Based Public Hazard Detection**” is studied and written by us under the guidance of **S.R. Tiwari**, Asst. Prof., Department of Artificial Intelligence and Data Science, Shri Sant Gadge Baba College of Engineering and Technology, Bhusawal.

We further declare that to the best of our knowledge, the project does not contain any part of the work which has been submitted for the award of any degree to either this university or any other. Such material as obtained from sources for study has been duly acknowledged and the name of the experts kept secret as per their request in data collection & analyzing for the project.

This report is written by studying various articles, books, papers, journals and other resources available on internet out of which some of them are listed in the end of report.

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## **ABSTRACT**

The AI Based Public Hazard Detection System is an intelligent real-time monitoring system developed using Artificial Intelligence, Deep Learning, and Computer Vision technologies. The purpose of this project is to detect hazardous road conditions such as potholes, road cracks, waterlogging, and damaged roads using live camera surveillance. The system uses the YOLOv8 object detection model integrated with OpenCV to analyze video frames captured through a webcam or CCTV camera. Whenever a hazard is detected, the system automatically highlights the affected area with bounding boxes and generates an alert sound to notify users immediately.

The proposed system helps improve public safety by reducing road accidents and enabling faster identification of hazardous conditions. Traditional manual inspection methods require significant time and manpower, whereas this AI-based system provides automatic and continuous monitoring with high accuracy and real-time performance. The project demonstrates the practical implementation of deep learning models for smart surveillance and smart city applications.

The system is trained on a custom dataset containing multiple road hazard images and performs real-time inference on live video streams. The project is cost-effective, scalable, and suitable for applications such as smart transportation systems, municipal road maintenance, traffic monitoring, and public safety management. This project highlights the growing importance of Artificial Intelligence in building safer and smarter urban environments.

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# CHAPTER 1

## INTRODUCTION

### 1.1) Introduction:

Urban infrastructure and public safety management have become major concerns in rapidly growing cities across the world. Increasing population density, traffic congestion, industrial growth, and environmental challenges continuously put pressure on public infrastructure systems such as roads, drainage networks, bridges, and public utilities. Hazards like potholes, waterlogging, garbage accumulation, damaged roads, open manholes, and accidents not only affect transportation and daily activities but also create serious risks to human life and property. Traditional methods of monitoring these hazards are mostly manual, time-consuming, and inefficient, making it difficult for authorities to respond quickly and effectively.

The advancement of Artificial Intelligence (AI), Computer Vision, and Smart Surveillance technologies has created new opportunities for developing intelligent urban safety systems capable of real-time monitoring and automated hazard detection. AI-powered systems can continuously analyze images and video streams from CCTV cameras, drones, and mobile devices to identify hazardous situations accurately and rapidly. Such intelligent systems reduce human effort, improve response time, and support data-driven decision-making for municipal authorities and smart city management.

The AI-Based Public Hazard Detection System is proposed as an advanced smart monitoring platform designed to detect, classify, and report multiple public hazards automatically using deep learning models such as YOLO and CNN. The system aims to improve urban safety, reduce accident risks, and assist authorities in maintaining public infrastructure more efficiently. By integrating AI technologies with GIS mapping, alert systems, and data analytics, the project contributes toward building safer, smarter, and more sustainable cities.

### 1.2) Background:

Public infrastructure plays a vital role in maintaining safety, transportation, communication, and daily activities in modern urban environments. Roads, bridges, footpaths, drainage systems, streetlights, and other public facilities are continuously used by millions of people every day. However, due to increasing urbanization, population growth, traffic congestion, and environmental factors, public infrastructure is frequently exposed to hazards such as potholes, road cracks, waterlogging, garbage accumulation, damaged streetlights, open manholes,

accidents, and structural failures. These hazards create serious risks for public safety and often result in accidents, vehicle damage, traffic delays, environmental pollution, and inconvenience to citizens.

Traditionally, public hazard monitoring is performed manually through field inspections or citizen complaints. Municipal authorities generally depend on workers to physically inspect roads and public places or rely on people to report hazards through calls or online complaint systems. Although these methods are widely used, they are slow, inefficient, labor-intensive, and highly dependent on human observation. Many hazards remain undetected for long periods, especially in crowded urban areas where continuous monitoring is difficult. Delayed detection and response may lead to severe consequences, including injuries, loss of life, economic losses, and reduced public trust in infrastructure management systems.

With the rapid advancement of technology, Artificial Intelligence (AI), Computer Vision, Machine Learning, and Deep Learning have emerged as powerful tools capable of solving complex real-world problems. AI systems can analyze large amounts of visual data from CCTV cameras, drones, mobile devices, and surveillance systems in real time. Using advanced deep learning algorithms such as Convolutional Neural Networks (CNN), YOLO (You Only Look Once), Faster R-CNN, and semantic segmentation models, it has become possible to automatically detect and classify various public hazards accurately and efficiently. These technologies reduce dependency on manual monitoring and provide intelligent automation for public safety management.

The AI-Based Public Hazard Detection System is designed to address these urban safety challenges by developing an automated hazard monitoring and reporting platform. The system captures images and video streams from different sources and processes them using AI-based object detection models. Hazards such as potholes, waterlogging, garbage dumps, accidents, crowd congestion, fire outbreaks, and damaged infrastructure are automatically identified and categorized according to severity levels. Once detected, alerts and notifications are generated for municipal authorities, allowing faster response and maintenance actions.

In addition to hazard detection, the system also supports GIS-based mapping and data analytics, helping authorities identify hazard-prone zones and analyze infrastructure conditions over time. Historical data storage and trend analysis enable predictive maintenance, allowing government departments to take preventive measures before hazards become critical. This approach supports the concept of Smart Cities, where automation, intelligent surveillance, and data-driven decision-making are used to improve urban management and public safety.

The proposed system aims to create a scalable, reliable, and efficient framework for real-time hazard monitoring. By integrating AI technologies with smart city infrastructure, the project contributes toward safer roads, improved municipal management, reduced accident rates, and better quality of life for citizens. The use of automated hazard detection systems can significantly transform the traditional public safety model into a modern intelligent monitoring ecosystem capable of supporting future urban development initiatives.

### **1.3) AI in Smart Cities:**

The concept of Smart Cities has gained significant importance in recent years due to rapid urbanization, technological advancements, and the increasing demand for efficient urban management. A Smart City uses modern technologies such as Artificial Intelligence (AI), Internet of Things (IoT), Cloud Computing, Big Data Analytics, Geographic Information Systems (GIS), and Computer Vision to improve the quality of public services, optimize infrastructure utilization, enhance environmental sustainability, and ensure public safety. The primary goal of smart city development is to create intelligent urban environments where resources are managed efficiently and citizens experience a safer, cleaner, and more connected lifestyle.

Artificial Intelligence plays a major role in transforming traditional cities into smart cities. AI enables machines and systems to analyze data, recognize patterns, make predictions, and perform intelligent decision-making without continuous human intervention. In smart city applications, AI is widely used in traffic management, healthcare systems, surveillance, waste management, disaster prediction, public transportation, environmental monitoring, and infrastructure maintenance. AI-based automation reduces manual work, increases efficiency, minimizes operational costs, and improves the speed and accuracy of decision-making processes.

One of the most important applications of AI in smart cities is intelligent surveillance and public safety monitoring. Urban environments generate massive amounts of visual data through CCTV cameras, drones, and mobile devices. Monitoring such large-scale visual information manually is practically impossible. AI-powered computer vision systems solve this problem by automatically analyzing live video streams and detecting abnormal events or hazards in real time. Technologies such as YOLO (You Only Look Once), Convolutional Neural Networks (CNN), Faster R-CNN, and image segmentation models are capable of identifying objects, accidents, road damages, crowd congestion, waterlogging, and suspicious activities with high accuracy.

AI-based systems also support predictive analytics and preventive maintenance in smart cities. By analyzing historical hazard data, traffic patterns, weather conditions, and infrastructure performance, AI models can predict future risks and identify areas prone to repeated failures. For example, roads that frequently develop potholes can be identified in advance, enabling authorities to take preventive action before major damage occurs. This predictive capability improves urban planning, reduces maintenance costs, and enhances public safety.

In addition, AI integrated with GIS and GPS technologies provides accurate location tracking and visualization of hazards on digital city maps. Municipal authorities can monitor city conditions through centralized dashboards, receive automated alerts, and deploy emergency response teams efficiently. AI-driven smart city systems also support sustainability by optimizing energy usage, reducing traffic congestion, improving waste management, and enabling efficient resource allocation.

The AI-Based Public Hazard Detection System directly supports smart city objectives by providing automated hazard detection, real-time monitoring, intelligent alerts, and data-driven decision support. The system enhances urban safety management by integrating AI, Computer Vision, GIS mapping, and cloud-based analytics into a unified framework. Through continuous surveillance and intelligent automation, the system contributes toward safer infrastructure, improved emergency response, and smarter governance.

As smart city initiatives continue to expand globally, AI technologies are expected to become even more advanced and deeply integrated into public infrastructure systems. Future smart cities will rely heavily on AI-driven automation, edge computing, IoT devices, and intelligent surveillance systems to manage urban environments more efficiently. Therefore, implementing AI-based public hazard detection systems is an important step toward building intelligent, sustainable, and future-ready cities.

#### **1.4) Public Safety Challenges:**

Public safety is one of the most important responsibilities of municipal authorities and urban management systems. Rapid urbanization, increasing population density, heavy traffic, environmental changes, and aging infrastructure have created several challenges related to maintaining safe public environments. Roads, bridges, drainage systems, public transport facilities, and utility infrastructures are continuously exposed to damage and hazardous conditions, which may lead to accidents, injuries, economic losses, and disruption of public services. Ensuring timely detection and management of such hazards remains a major challenge for modern cities.

One of the most common public safety challenges is road infrastructure damage. Potholes, road cracks, uneven surfaces, damaged footpaths, and open manholes are frequently observed in urban and semi-urban areas. These hazards increase the risk of vehicle accidents, traffic congestion, and pedestrian injuries. During rainy seasons, waterlogging and flooding further worsen road conditions and create dangerous situations for commuters. In many cases, poor drainage systems and delayed maintenance result in long-term infrastructure deterioration.

Another major challenge is the lack of continuous monitoring and real-time hazard detection. Most municipal systems still rely on manual inspections or citizen complaints to identify public hazards. This traditional approach is slow, reactive, and inefficient because hazards are usually reported only after significant damage or accidents occur. Large urban areas contain thousands of roads and public spaces, making it difficult for authorities to monitor every location manually. As a result, many hazards remain unnoticed for extended periods.

Public waste management and environmental hazards also pose serious safety concerns. Garbage accumulation on roadsides, illegal dumping, overflowing waste bins, and poor sanitation conditions contribute to pollution, spread of diseases, and reduced quality of life. In densely populated areas, improper waste management may also block drainage systems and increase flood risks during heavy rainfall. Detecting and managing such environmental hazards efficiently is essential for maintaining urban hygiene and sustainability.

Crowd management and accident detection are additional challenges faced in modern urban environments. Public gatherings, traffic congestion, road accidents, and emergency situations require immediate response from authorities. Manual surveillance through CCTV cameras is difficult because human operators cannot continuously monitor multiple video streams effectively. Fatigue, distraction, and limited manpower often result in delayed responses to emergencies. Real-time AI-based surveillance systems can help overcome these limitations by automatically identifying abnormal activities, accidents, or crowd anomalies.

Fire outbreaks, structural damage, and disaster-related incidents are also critical public safety issues. Electrical failures, industrial accidents, gas leaks, building collapses, and natural disasters can occur suddenly and require immediate action. Early detection and rapid response are essential to minimize damage and save lives. However, conventional monitoring systems lack intelligent automation and predictive capabilities needed for effective emergency management.

Another important challenge is the absence of centralized hazard management and data analysis systems. Most existing systems do not maintain structured historical records of hazards, making it difficult to analyze trends, identify high-risk zones, or plan preventive

maintenance strategies. Without proper data analytics and visualization tools, authorities cannot make informed decisions regarding infrastructure planning and resource allocation.

The AI-Based Public Hazard Detection System aims to address these public safety challenges through intelligent automation, real-time surveillance, AI-based detection, GIS mapping, and data analytics. By reducing dependency on manual monitoring and enabling faster response mechanisms, the proposed system contributes toward safer public infrastructure, efficient urban management, and improved quality of life for citizens.

### **1.5) Need for Automation:**

With the rapid growth of urban populations and increasing pressure on public infrastructure, traditional methods of hazard monitoring are no longer sufficient to ensure public safety and efficient infrastructure management. Modern cities require intelligent systems capable of continuous monitoring, quick decision-making, and real-time hazard detection. Automation has therefore become an essential requirement for managing public hazards effectively and supporting smart city development initiatives.

Public hazards such as potholes, road cracks, garbage accumulation, waterlogging, open manholes, accidents, fire outbreaks, and damaged infrastructure occur frequently in urban environments. Detecting these hazards manually is difficult because cities contain large road networks, public spaces, and infrastructure facilities spread across wide geographical areas. Human inspection teams cannot continuously monitor every location, especially during night hours, heavy traffic conditions, or adverse weather situations. As a result, many hazards remain undetected for long periods, increasing the risk of accidents and public inconvenience.

Automation using Artificial Intelligence (AI), Computer Vision, and Deep Learning provides a reliable solution to these challenges. AI-based systems can continuously analyze images and video feeds from CCTV cameras, drones, and mobile devices without interruption. Unlike human operators, automated systems can function 24/7 without fatigue, distraction, or inconsistency. This ensures continuous surveillance and immediate detection of hazardous situations.

Another important reason for automation is the need for faster response time. In manual systems, authorities often receive hazard information only after citizens report the issue or after scheduled inspections are completed. This delay may result in severe consequences such as road accidents, traffic congestion, environmental pollution, or infrastructure failure. Automated hazard detection systems instantly identify hazards and generate alerts, enabling authorities to take corrective action quickly and reduce public risks.



Automation also improves accuracy and efficiency in hazard detection. Advanced AI models such as YOLO, CNN, and Faster R-CNN are capable of detecting multiple hazard types with high precision under different environmental conditions. These systems can recognize patterns, classify hazards, estimate severity levels, and provide intelligent recommendations for maintenance and emergency response. This reduces human error and improves the reliability of public safety systems.

In addition, automated systems support centralized data storage, historical analysis, and predictive maintenance. All detected hazards can be stored in databases and analyzed using data analytics tools to identify trends and high-risk areas. Authorities can use this information for long-term planning, infrastructure improvement, and smart resource allocation. Automation therefore not only improves hazard detection but also enhances overall urban management and decision-making processes.

The AI-Based Public Hazard Detection System addresses the growing need for automation by integrating AI technologies, real-time surveillance, GIS mapping, and intelligent analytics into a unified platform. The system contributes toward safer roads, faster hazard management, reduced operational costs, and improved smart city infrastructure management.

#### **1.6) Existing Manual System Problems:**

The existing public hazard monitoring systems used by municipal authorities mainly depend on manual inspection methods, citizen complaints, and conventional surveillance techniques. Although these approaches have been used for many years, they suffer from several limitations that reduce their efficiency, reliability, and scalability in modern urban environments.

One of the major problems with manual systems is delayed hazard detection. Municipal workers usually conduct periodic inspections of roads and public areas, which means hazards may remain unnoticed between inspection intervals. Citizens also report hazards only after they become severe or cause inconvenience. Due to these delays, authorities often fail to take timely corrective action, increasing the risk of accidents and infrastructure damage.

Manual monitoring systems are also highly labor-intensive and expensive. Large urban areas contain thousands of roads, drainage systems, public spaces, and infrastructure facilities that require continuous inspection. Maintaining sufficient manpower for regular monitoring is difficult and costly for municipal authorities. Furthermore, manual inspections consume significant time and resources, reducing operational efficiency.

Human error is another major limitation of traditional systems. Continuous monitoring of CCTV footage or public spaces is difficult because human operators may become tired,

distracted, or inconsistent during long working hours. Important hazards may therefore be missed or incorrectly classified. Environmental factors such as low lighting, rain, fog, or traffic congestion further reduce the effectiveness of manual surveillance systems.

Another problem is the lack of real-time monitoring and automated alerts. Existing systems generally do not provide instant hazard detection or immediate notifications to authorities. As a result, emergency situations such as accidents, fire outbreaks, or flooding may not receive quick attention. Delayed response times increase public risk and may lead to greater infrastructure damage or loss of life.

Traditional systems also lack centralized data management and intelligent analytics capabilities. Most hazard reports are stored manually or in isolated databases, making it difficult to analyze trends, identify recurring problems, or perform predictive maintenance. Authorities therefore struggle to make data-driven decisions regarding infrastructure planning and resource allocation.

In addition, existing systems often focus only on specific types of hazards and fail to provide a unified monitoring solution. Separate systems may be used for traffic management, waste management, road inspection, or emergency response, leading to fragmented operations and poor coordination between departments.

The absence of GIS-based mapping and real-time visualization is another significant limitation. Without accurate geographic information, authorities may face difficulties in locating hazards quickly or identifying high-risk zones within the city. This reduces the efficiency of maintenance and emergency response teams.

These limitations highlight the need for an intelligent, automated, and integrated hazard detection system capable of continuous monitoring, real-time alerts, AI-based analysis, and centralized management.

### **1.7) Objectives of the Project:**

The main objective of the AI-Based Public Hazard Detection System is to develop an intelligent and automated platform capable of detecting, monitoring, and reporting public hazards in real time using Artificial Intelligence, Computer Vision, and Deep Learning technologies. The system aims to improve public safety, reduce manual monitoring efforts, and support smart city infrastructure management through intelligent automation and data-driven decision-making.

The specific objectives of the project are as follows:

### **1. To Develop an Automated Hazard Detection System**

The primary objective of the project is to create an AI-powered system capable of automatically detecting hazards from images and video streams without continuous human supervision. The system will analyze CCTV footage, drone visuals, and uploaded images to identify hazardous conditions efficiently.

### **2. To Detect Multiple Types of Public Hazards**

The system is designed to detect various urban hazards such as potholes, road cracks, waterlogging, garbage dumps, accidents, crowd congestion, open manholes, and damaged infrastructure. Detecting multiple hazards within a single platform improves overall urban safety monitoring.

### **3. To Use AI and Deep Learning Techniques**

Another important objective is to implement advanced Artificial Intelligence and Deep Learning algorithms such as YOLO, CNN, and Faster R-CNN for accurate hazard detection and classification. These models help improve detection speed, accuracy, and real-time performance.

### **4. To Reduce Manual Monitoring Efforts**

The project aims to minimize dependency on manual inspections and human surveillance systems. Automated hazard detection reduces labor costs, improves efficiency, and enables continuous monitoring of public infrastructure.

### **5. To Generate Real-Time Alerts and Notifications**

The system is intended to provide immediate alerts and notifications whenever a hazard is detected. This allows municipal authorities and emergency teams to respond quickly and prevent accidents or infrastructure failures.

### **6. To Implement Severity Analysis of Hazards**

The project also aims to classify detected hazards into different severity levels such as Low, Medium, and High risk. Severity-based categorization helps authorities prioritize urgent maintenance activities and allocate resources effectively.

### **7. To Integrate GIS-Based Hazard Mapping**

The system integrates Geographic Information System (GIS) and GPS technologies to map hazards accurately on digital city maps. This helps authorities identify hazard-prone zones and improve infrastructure planning and management.

### **8. To Store Hazard Data for Future Analysis**

All detected hazards, including images, timestamps, location details, and severity levels, will be stored in a centralized database. This historical data can be used for trend analysis, predictive maintenance, and future urban planning.

### **9. To Support Smart City Development**

The project supports smart city initiatives by introducing intelligent automation, real-time monitoring, and AI-based surveillance systems. It contributes toward safer, cleaner, and more sustainable urban environments.

### **10. To Develop a Scalable and Future-Ready System**

The proposed system is designed to be scalable and adaptable for future technologies such as IoT sensors, cloud computing, mobile applications, and emergency response integration. This ensures long-term usability and expansion capabilities for large-scale smart city deployment.

## **CHAPTER 2**

### **LITERATURE REVIEW**

The development of intelligent public hazard detection systems has become an important research area due to rapid urbanization, increasing traffic density, and the growing need for safer public infrastructure. Traditional hazard monitoring methods are no longer sufficient for managing large-scale urban environments because they rely heavily on manual inspection and delayed reporting mechanisms. Recent advancements in Artificial Intelligence (AI), Computer Vision, Deep Learning, Internet of Things (IoT), and Smart Surveillance Systems have enabled the development of automated hazard detection solutions capable of real-time monitoring and intelligent decision-making.

This chapter presents the historical evolution of hazard detection systems, reviews existing literature and research contributions related to AI-based public hazard detection, and identifies the research gaps that motivate the proposed system. The chapter also discusses the problem definition addressed by this project.

#### **2.1 History:**

The history of Public Hazard Detection Systems traces its development from traditional manual monitoring methods to modern AI-driven smart surveillance technologies. Over the years, advancements in Artificial Intelligence (AI), Computer Vision, Deep Learning, IoT, and Smart City infrastructure have significantly transformed the way hazards are detected, monitored, and managed. Here's a brief overview:

- **Early Foundations (1900s–1960s)**

**1900s–1930s:** During the early years, public hazard monitoring was completely manual. Municipal workers and local authorities physically inspected roads, bridges, drainage systems, and public spaces to identify hazards such as damaged roads, floods, fire outbreaks, and structural failures. Hazard reporting mainly depended on citizen complaints and field inspections.

**1940s–1950s:** Governments and urban authorities started implementing organized public safety departments for disaster response and infrastructure maintenance. Basic alarm systems and communication technologies were introduced for emergency reporting and coordination.

**1960s:** Closed-Circuit Television (CCTV) systems were introduced for surveillance in public places such as railway stations, airports, and traffic intersections. Although these systems

improved monitoring capabilities, they still required continuous human observation and manual analysis.

- **Growth of Digital Surveillance (1970s–1990s)**

**1970s:** Urbanization and increasing traffic created a greater need for intelligent public safety systems. Traffic monitoring cameras and electronic surveillance systems became more common in developed cities. Authorities started using digital recording systems for incident tracking and security management.

**1980s:** Researchers introduced early image processing techniques for object recognition and infrastructure analysis. Basic computer vision algorithms such as edge detection, pattern recognition, and image filtering were used for identifying road damages and abnormal events. However, these systems had limited accuracy and struggled under real-world environmental conditions.

**1990s:** With the growth of computer technology and the internet, digital image processing advanced rapidly. Feature extraction methods such as SIFT (Scale Invariant Feature Transform), SURF (Speeded Up Robust Features), and HOG (Histogram of Oriented Gradients) were developed for automated visual analysis. Public hazard monitoring systems started shifting from purely manual approaches toward semi-automated solutions.

- **Rise of Artificial Intelligence and Machine Learning (2000s–2010s)**

**2000s:** Artificial Intelligence (AI) and Machine Learning (ML) began transforming surveillance and hazard detection systems. Researchers developed machine learning models capable of recognizing patterns in images and videos. Online databases, cloud computing, and digital surveillance systems improved the collection and storage of hazard-related data.

**Mid-2000s:** Convolutional Neural Networks (CNNs) emerged as powerful tools for image classification and object recognition. These models improved the accuracy of detecting road cracks, potholes, accidents, and infrastructure damage from images.

**2010s:** Deep Learning technologies revolutionized public hazard detection systems. Advanced object detection models such as YOLO (You Only Look Once), Faster R-CNN, SSD (Single Shot Detector), and Mask R-CNN enabled real-time detection of hazards from CCTV feeds and street-level images. AI-based systems became capable of identifying potholes, waterlogging, garbage accumulation, crowd congestion, accidents, and fire hazards with high accuracy.

- **Modern Smart City Era (2015–Present)**

**2015–2020:** Smart City initiatives accelerated the integration of AI, IoT sensors, GIS mapping, cloud computing, and drone technology into public safety systems. Governments and

researchers focused on building intelligent urban monitoring platforms capable of real-time hazard detection and automated alert generation. GIS-based mapping allowed authorities to locate hazards accurately and identify high-risk zones within cities.

**2020–Present:** Modern public hazard detection systems now use advanced AI models, edge computing, real-time analytics, and cloud-based dashboards for continuous monitoring and decision-making. Deep learning algorithms analyze data from CCTV cameras, drones, mobile devices, and IoT sensors to automatically detect hazards and classify severity levels. Predictive maintenance and trend analysis are also used to forecast future infrastructure failures and optimize urban planning.

#### • **Present and Future Trends**

Today, AI-Based Public Hazard Detection Systems are becoming an important part of Smart City infrastructure. These systems support automated surveillance, intelligent transportation management, environmental monitoring, and emergency response coordination. Modern systems can detect multiple hazards simultaneously, generate real-time alerts, store historical data, and provide decision support for municipal authorities.

Future developments are expected to include integration with autonomous drones, IoT-based smart sensors, 5G communication networks, edge AI devices, and fully automated emergency response systems. These advancements will further improve urban safety, reduce response time, and contribute toward safer, smarter, and more sustainable cities.

## **2.2 Literature**

Several researchers and organizations have contributed to the development of AI-based hazard detection systems using Computer Vision, Deep Learning, GIS mapping, and smart surveillance technologies. A review of existing literature helps understand current methodologies, strengths, limitations, and research gaps related to public hazard detection systems.

### **1. Real-Time Pothole Detection using Deep Learning**

Al-Shaghouri, Alkhatib, and Berjaoui (2021) proposed a real-time pothole detection system using Convolutional Neural Networks (CNN). Their system analyzed street-level images to identify potholes automatically and improve road safety monitoring. The study demonstrated that deep learning models provide better accuracy compared to traditional image processing methods. However, the system mainly focused only on pothole detection and lacked integration with real-time alert systems and GIS mapping.

## **2. Vision-Based Road Hazard Detection Systems**

A review paper published in *Sensors Journal* (2024) analyzed various deep learning techniques used for automated road hazard detection. The paper discussed CNN, YOLO, Faster R-CNN, semantic segmentation, and object detection models used for detecting potholes, road cracks, and damaged infrastructure. The research highlighted that YOLO-based systems achieved faster real-time performance, making them suitable for smart city applications. However, the paper also identified challenges such as environmental interference, dataset limitations, and computational requirements.

## **3. Disaster-Based Road Damage Detection**

Chakraborty, Lee, and Kim (2025) developed a deep learning framework for detecting road damage caused by natural disasters using satellite imagery. Their research focused on rapid identification of damaged roads after earthquakes and floods to support emergency response systems. The study showed how AI and satellite-based monitoring could improve disaster management efficiency. However, the approach required high-resolution satellite images and powerful computational infrastructure.

## **4. Transfer Learning for Pothole Detection**

Parasnis, Chokshi, Jain, and Devadkar (2023) introduced RoadScan, a transfer learning-based pothole detection framework using YOLO models. Their approach improved detection accuracy by using pre-trained deep learning models and road surface image datasets. The system was capable of real-time pothole detection with reduced training time. Although the framework showed promising results, it focused primarily on road hazards and did not include multi-hazard detection capabilities.

## **5. Road Damage Dataset for Deep Learning**

Arya, Maeda, Ghosh, Toshniwal, and Sekimoto (2021) introduced the RDD2020 dataset for automatic road damage detection. The dataset included annotated images of road cracks, potholes, and surface damages collected from multiple countries. Researchers evaluated different deep learning models such as YOLO and EfficientDet using this dataset. The study contributed significantly toward improving training data availability for hazard detection research.

## **6. AI-Based Urban Surveillance Systems**

Several recent studies have focused on integrating AI surveillance with smart city infrastructure. Researchers have proposed systems capable of detecting crowd congestion, accidents, abandoned objects, and public safety threats using CCTV footage and Computer



Vision models. These systems improved urban monitoring efficiency but often lacked centralized management, predictive analytics, and scalable deployment strategies.

## **7. GIS and IoT Integration in Hazard Monitoring**

Modern research also emphasizes the use of GIS mapping, GPS tracking, IoT sensors, and cloud computing for smart hazard management. GIS-based systems enable authorities to visualize hazards geographically and identify high-risk zones. IoT sensors improve environmental monitoring by detecting water levels, temperature, gas leaks, and fire risks. Although these technologies improve monitoring capabilities, integrating them into a unified intelligent platform remains a challenge.

The literature review shows that existing systems provide valuable contributions toward automated hazard detection, but many solutions remain limited to specific hazard categories or isolated technologies. Most current systems lack complete integration of real-time detection, AI-based analytics, GIS mapping, centralized dashboards, predictive maintenance, and emergency response mechanisms. Therefore, there is a need for a comprehensive AI-Based Public Hazard Detection System capable of providing scalable, intelligent, and real-time urban safety management.

### **2.3 Problem Definition**

Urban environments face multiple public hazards such as potholes, road cracks, waterlogging, garbage accumulation, open manholes, damaged infrastructure, traffic accidents, fire outbreaks, and crowd congestion. These hazards negatively impact public safety, transportation systems, environmental conditions, and overall quality of life. Delayed identification and maintenance of such hazards may lead to accidents, injuries, traffic delays, infrastructure damage, economic losses, and environmental pollution.

Currently, most municipal authorities depend on manual inspections, citizen complaints, and conventional surveillance systems to detect public hazards. These traditional methods are inefficient, slow, labor-intensive, and highly dependent on human observation. Since continuous monitoring of large urban areas is difficult, many hazards remain unnoticed for extended periods. Existing systems also lack real-time monitoring, intelligent automation, predictive analysis, and centralized hazard management capabilities.

Human operators monitoring CCTV feeds may fail to identify hazards due to fatigue, distraction, or limited attention span. Furthermore, separate systems used by different departments create fragmented operations and reduce coordination efficiency. Most current

solutions focus only on single hazard categories such as potholes or accidents and fail to provide an integrated multi-hazard monitoring framework.

With increasing urbanization and smart city development, there is a strong requirement for an intelligent AI-based automated hazard detection system capable of continuous monitoring, real-time detection, hazard classification, GIS-based location tracking, automated alerts, and data-driven decision-making.

Therefore, the main problem addressed by this project is the lack of a unified intelligent system that can automatically detect multiple public hazards from images and video feeds, analyze their severity, generate real-time alerts, and assist municipal authorities in improving public safety and urban infrastructure management through AI-based automation.

## CHAPTER 3

### SYSTEM ANALYSIS & DESIGN

#### 3.1 System Architecture

The system architecture represents the overall structure and workflow of the AI-Based Public Hazard Detection System. The architecture explains how video frames are captured, processed, analyzed, and displayed by the system.

The proposed system begins with a webcam or CCTV camera that continuously captures real-time video frames. These frames are sent to the preprocessing module, where image resizing and normalization operations are performed to improve detection accuracy and processing speed.

After preprocessing, the frames are passed to the AI detection layer. YOLO is used for detecting persons and analyzing crowd conditions by counting the number of detected people. Fall detection is performed by analyzing body posture and object dimensions, while MediaPipe is used to identify SOS emergency gestures based on body landmark detection.

Whenever the system detects a crowd, fallen person, or SOS gesture, an alert message is generated automatically. The final output with bounding boxes, labels, and warning messages is displayed on the monitoring screen in real time.

The architecture provides continuous monitoring, fast detection speed, and automated emergency identification for public safety applications.

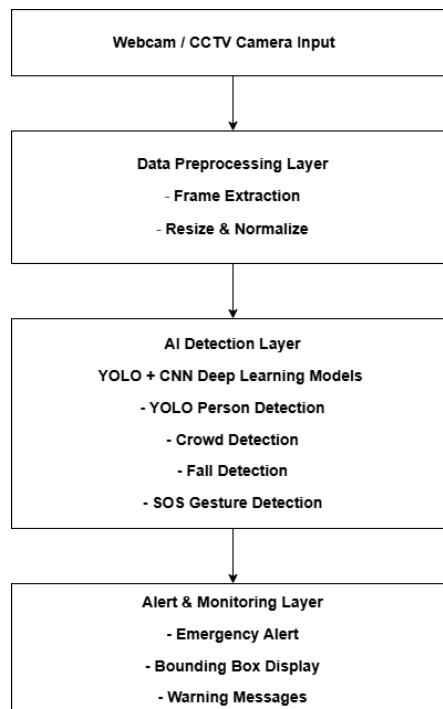


Fig 3.1: System Architecture Diagram

### 3.2 Flowchart

The flowchart represents the logical flow of the AI-Based Public Hazard Detection System. It describes the sequence of operations performed during the hazard detection process.

The system starts by capturing video frames from the camera. The captured frames are preprocessed and then passed to the YOLO model for person detection. After detecting persons, the system performs crowd detection by counting the number of people present in the frame. Next, fall detection is performed by analyzing the posture and dimensions of detected persons. The system then checks for SOS emergency gestures using MediaPipe landmark detection techniques. If any hazard or emergency situation is identified, an alert message is generated and displayed on the screen. Finally, the processed output is displayed continuously until the system stops.

The flowchart helps in understanding the overall working process and decision flow of the proposed system.

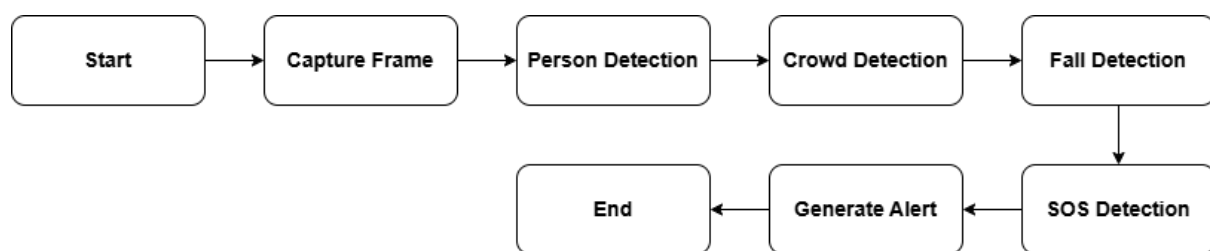


Fig 3.2: Flow Chart

### 3.3 Activity Diagram

The activity diagram explains the operational workflow and activities performed by the AI-Based Public Hazard Detection System. It represents the sequence of actions from video capture to final output generation.

The process begins when the camera or webcam starts capturing live video frames. The frames are preprocessed to improve image quality and prepare them for AI analysis. The YOLO model performs person detection and identifies the number of people present in the frame.

The system then performs crowd detection and checks whether the number of detected people exceeds the defined threshold. Fall detection is carried out by analyzing body orientation and posture. MediaPipe is further used to detect SOS emergency gestures.

If any emergency condition such as crowd formation, fallen person, or SOS gesture is detected, the system generates warning alerts and displays the detection results on the monitoring screen. The activities continue repeatedly for real-time monitoring and hazard detection.

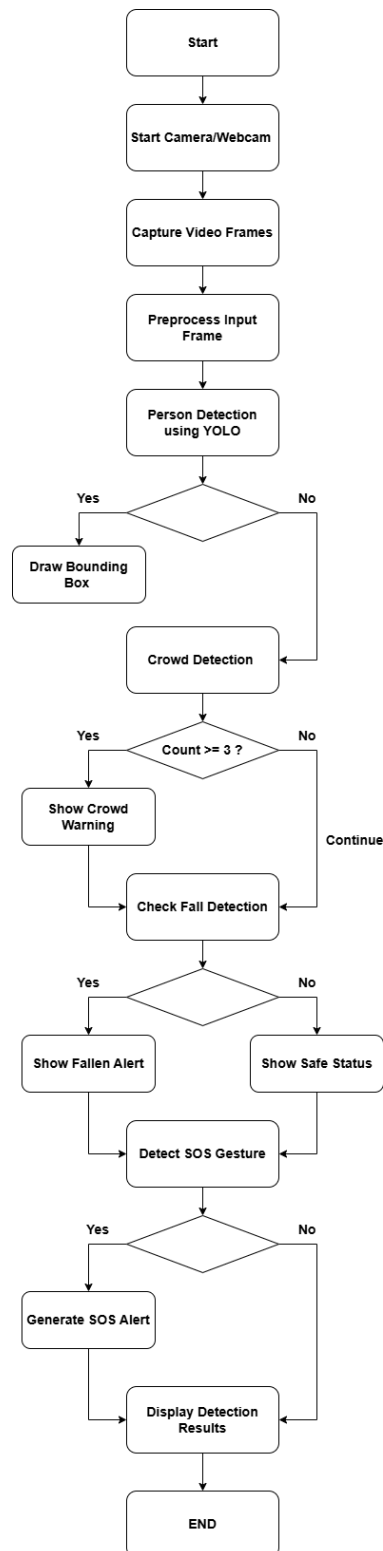


Fig 3.3: Activity Diagram

### 3.4 Sequence Diagram

The sequence diagram represents the interaction and communication between different modules of the AI-Based Public Hazard Detection System. It explains how data flows between the camera, preprocessing module, AI detection system, and output display.

Initially, the webcam or CCTV camera captures video frames and sends them to the preprocessing module. The preprocessing module prepares the frames and forwards them to the YOLO and MediaPipe models for analysis.

The AI detection system performs person detection, crowd detection, fall detection, and SOS gesture recognition. After processing, the detection results are sent to the alert generation module. If any hazard or emergency situation is identified, warning messages are generated automatically. Finally, the output with labels, alerts, and bounding boxes is displayed on the monitoring screen.

The sequence diagram helps in understanding communication flow and synchronization between different system components.

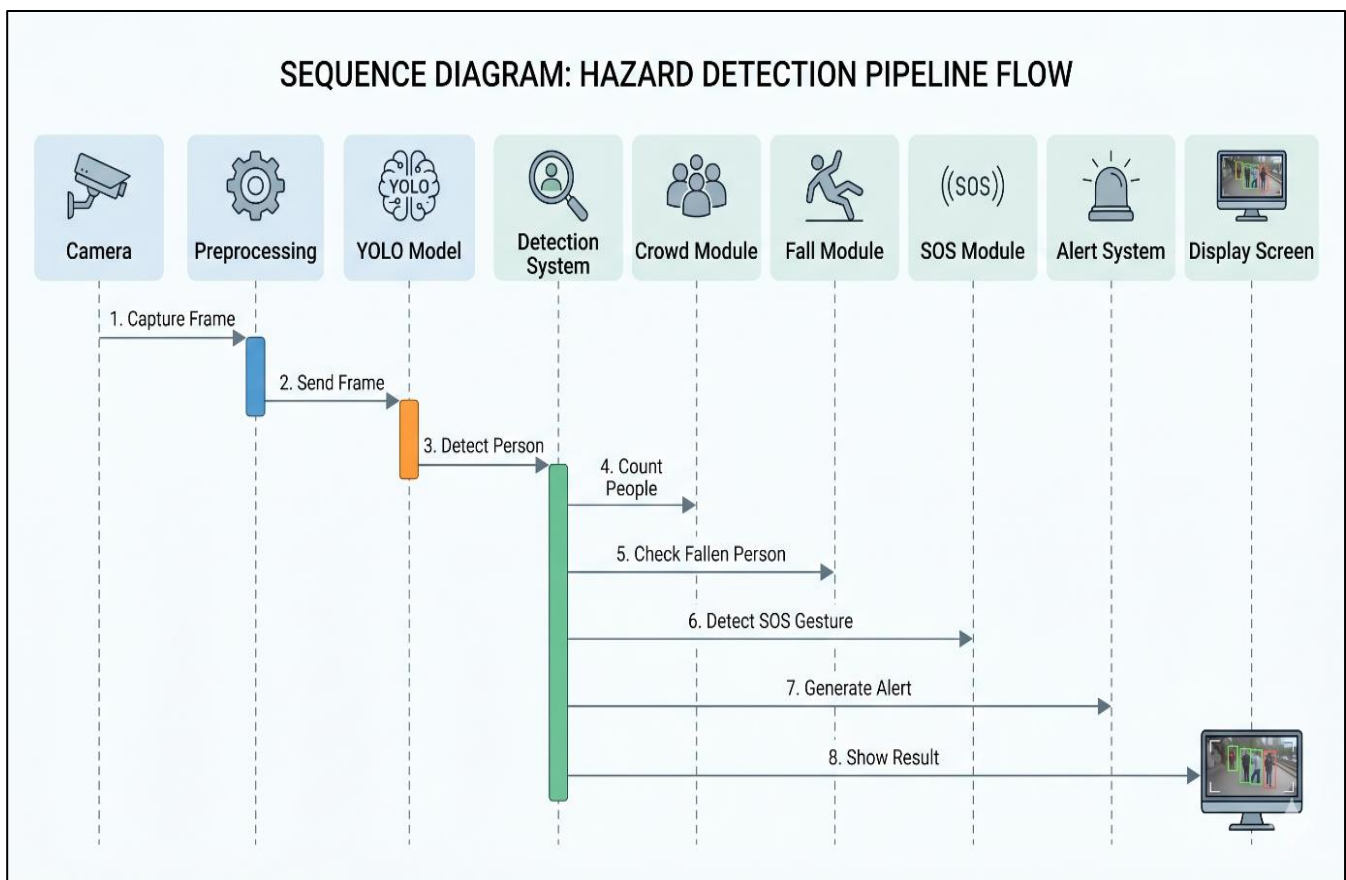


Fig 3.4: Sequence Diagram

### **3.5 AI Workflow**

The AI workflow represents the complete processing pipeline of the AI-Based Public Hazard Detection System. The workflow explains how the system processes live video frames and performs intelligent hazard detection using Artificial Intelligence and Computer Vision techniques.

The workflow begins with capturing real-time video input from a webcam or CCTV camera. The captured frames are sent to the preprocessing stage, where resizing, normalization, and frame enhancement operations are performed to improve detection accuracy and processing speed.

After preprocessing, the YOLO detection model is applied for person detection. The model identifies humans present in the video frame and generates bounding boxes around detected persons. Based on the number of detected individuals, the system performs crowd detection and identifies crowd formation conditions.

Further analysis is performed for fall detection using posture and body orientation analysis. The system checks whether the detected person is in a fallen position and generates emergency alerts if required. MediaPipe-based landmark detection is then used for SOS gesture recognition to identify emergency hand gestures or body movements.

Whenever the system detects crowd formation, fallen persons, or SOS emergency gestures, alert messages are generated automatically and displayed on the monitoring screen. The final output includes real-time detection labels, warning messages, and visual monitoring support.

The AI workflow enables automated monitoring, fast hazard detection, and intelligent public safety analysis for real-time surveillance applications.

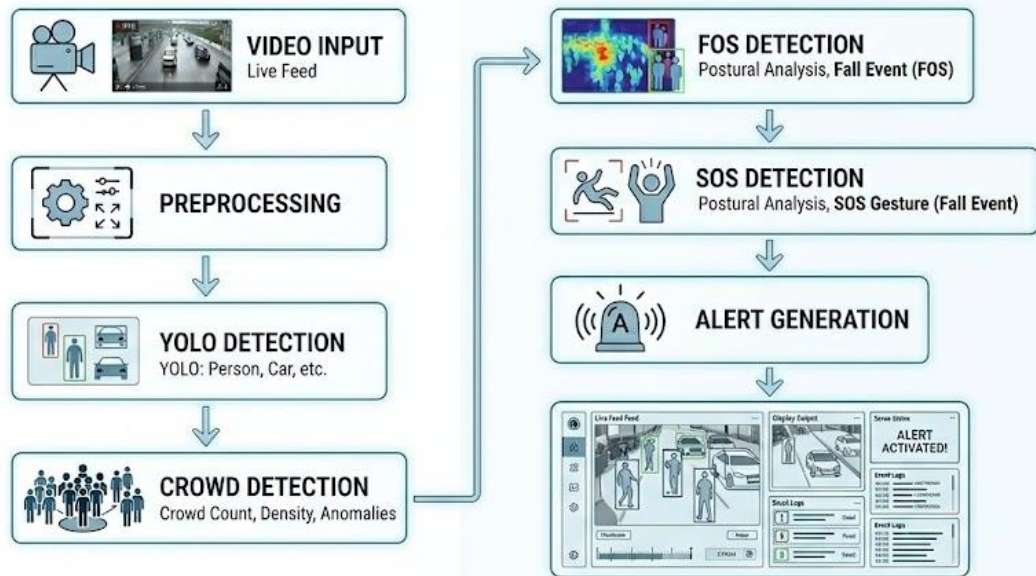


Fig 3.5: AI Workflow



## **CHAPTER 4**

### **IMPLEMENTATION & METHODOLOGY**

#### **4.1 Technologies Used**

The AI-Based Public Hazard Detection System uses several modern technologies and software libraries for implementation. These technologies help in real-time video processing, AI model execution, and hazard detection.

##### **Python**

Python is the main programming language used for implementing the proposed system. Python provides simple syntax, extensive AI libraries, and strong support for Computer Vision and Deep Learning applications. It is widely used in AI-based projects because of its flexibility and large developer community.

##### **OpenCV**

OpenCV (Open Source Computer Vision Library) is used for image and video processing. It helps in capturing live video frames, resizing images, drawing bounding boxes, displaying labels, and processing real-time camera streams. OpenCV plays an important role in frame manipulation and visualization.

##### **YOLOv8**

YOLOv8 (You Only Look Once Version 8) is a real-time object detection model used for person detection and crowd analysis. The model processes video frames and identifies humans with high speed and accuracy. YOLOv8 generates bounding boxes around detected persons and helps in crowd counting.

##### **MediaPipe**

MediaPipe is used for body posture analysis and SOS gesture detection. It detects body landmarks and hand movements, which helps the system identify emergency gestures and fallen positions. MediaPipe provides efficient real-time landmark tracking for human pose estimation.

##### **NumPy**

NumPy is a Python library used for numerical operations and array processing. It supports image matrix handling and helps improve processing efficiency during frame manipulation.

#### **4.2 Hardware Requirements**

The proposed system requires basic hardware components for real-time monitoring and AI-based detection.

| Hardware Component | Specification                    |
|--------------------|----------------------------------|
| Processor          | Intel i5 / Ryzen 5 or higher     |
| RAM                | Minimum 8 GB                     |
| Storage            | 256 GB SSD or higher             |
| Webcam/CCTV        | HD Camera                        |
| GPU (Optional)     | NVIDIA GPU for faster processing |
| Monitor            | Standard Display Monitor         |

Table 4.2: Hardware Requirements

The hardware configuration supports smooth execution of AI models and real-time video processing.

#### 4.3 Software Requirements

The proposed system uses different software tools and libraries for development and execution.

| Software          | Purpose                    |
|-------------------|----------------------------|
| Python            | Programming Language       |
| OpenCV            | Image and Video Processing |
| YOLOv8            | Person Detection           |
| MediaPipe         | Pose and Gesture Detection |
| NumPy             | Numerical Processing       |
| VS Code / PyCharm | Development Environment    |
| Windows/Linux     | Operating System           |

Table 4.3: Software Requirements

#### 4.4 Dataset Collection

Dataset collection is an important step in developing the AI-Based Public Hazard Detection System. The proposed system uses datasets related to person detection, crowd detection, fall detection, and SOS gesture recognition.

The datasets were collected from multiple sources including publicly available datasets from [Kaggle](#) and real-time video samples captured using webcams and CCTV cameras. Kaggle datasets provided images and videos for person detection, crowd analysis, and fall detection, while additional real-time samples were collected to improve practical testing and system performance.

The collected data included different lighting conditions, human postures, crowd situations, and emergency gesture variations to improve the robustness of the AI models. The datasets were further divided into training and testing samples for evaluating detection accuracy and system performance.

Real-time captured samples helped in testing the system under practical surveillance conditions and improved the reliability of hazard detection in real-world environments.



Fig 4.4.1: Person Detection Dataset Image 1



Fig 4.4.2: Person Detection Dataset Image 2



Fig 4.4.3: Crowd Detection Dataset Image 1



Fig 4.4.4: Crowd Detection Dataset Image 2



Fig 4.4.5: Fallen Person Dataset Image 1





Fig 4.4.6: Fallen Person Dataset Image 2



Fig 4.4.7: SOS Gesture Dataset Image 1



Fig 4.4.8: SOS Gesture Dataset Image 2

#### **4.5 Data Preprocessing**

Data preprocessing is performed to improve image quality and optimize the input data for AI model analysis. Preprocessing helps increase detection accuracy and reduce computational complexity.

The preprocessing stage includes image resizing, normalization, frame extraction, and noise reduction. Captured video frames are resized into fixed dimensions before being sent to the AI model. Normalization improves image consistency under different lighting conditions.

OpenCV functions are used for frame extraction and image processing operations. Proper preprocessing improves YOLO detection performance and enhances real-time processing speed.

#### 4.6 YOLOv8 Person Detection

YOLOv8 is used as the primary object detection model in the proposed system. The model detects humans present in real-time video frames and generates bounding boxes around detected persons.

YOLOv8 works using a single-stage object detection approach, where image classification and localization are performed simultaneously. This allows fast processing speed and accurate real-time detection.

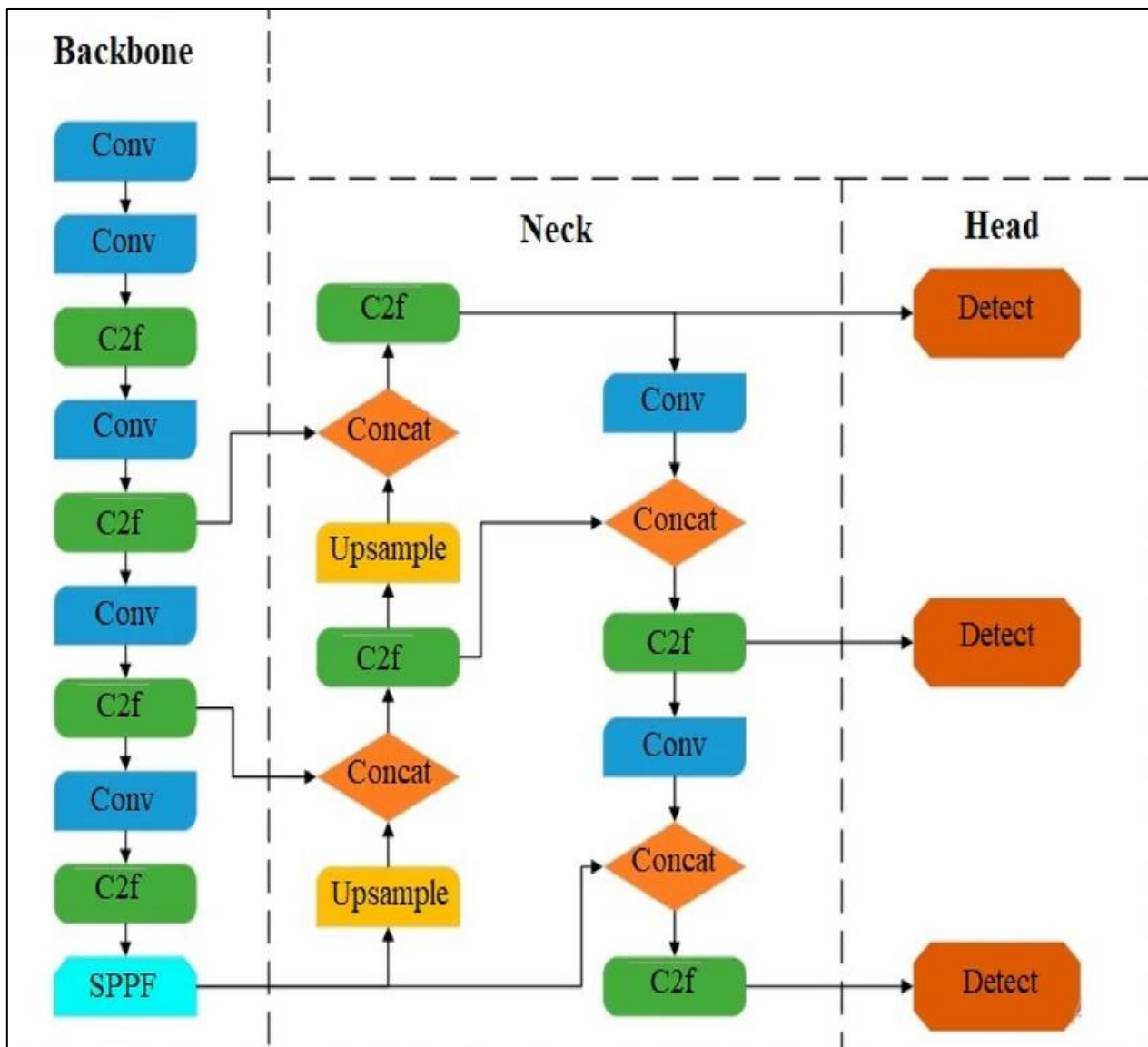


Fig 4.6: YOLOv8 architecture



The model continuously processes frames captured from the webcam or CCTV camera. Whenever a person is identified, a bounding box and label are displayed on the screen. The detected persons are further used for crowd analysis and fall detection.

YOLOv8 provides high detection accuracy and low latency, making it suitable for smart surveillance and public safety applications.

#### **4.7 Crowd Detection Method**

Crowd detection is performed based on the number of persons detected in the video frame. After YOLOv8 identifies individuals, the system counts the total number of detected persons. If the number of people exceeds a predefined threshold value, the system identifies the condition as crowd formation. A warning message is then generated automatically and displayed on the monitoring screen.

The crowd detection method helps in identifying overcrowded public areas and supports real-time public safety monitoring.

#### **4.8 Fall Detection Method**

Fall detection is an important feature of the proposed system. The system analyzes body posture and orientation to determine whether a person has fallen.

The detection process compares body width and height dimensions after person detection. If the detected body posture appears horizontal or abnormal for a specific duration, the system classifies it as a fallen condition.

Whenever a fallen person is detected, the system generates an emergency alert message. This feature is useful for accident monitoring and emergency response systems.

#### **4.9 SOS Gesture Detection**

SOS gesture detection is implemented using MediaPipe body landmark detection. MediaPipe tracks human body points and identifies specific emergency gestures or hand movements.

The system continuously analyzes body posture and gesture patterns. If an SOS gesture is recognized, the system immediately generates an emergency warning message on the monitoring screen.

SOS gesture detection improves emergency communication and allows individuals to request help during dangerous situations.

#### **4.10 Alert Generation System**

The alert generation system is responsible for displaying emergency notifications whenever hazardous situations are detected.

The system generates alerts for:

- Crowd Detection
- Fallen Person Detection
- SOS Gesture Detection

Alert messages are displayed on the screen in real time along with detection labels and bounding boxes. The alert generation system improves emergency response and supports intelligent surveillance monitoring.

#### **4.11 System Implementation Steps**

The implementation of the proposed system was carried out in multiple stages.

##### **Step 1: Install Required Libraries**

Python libraries such as OpenCV, YOLOv8, MediaPipe, and NumPy were installed.

##### **Step 2: Capture Video Input**

The webcam or CCTV camera was connected to capture live video streams.

##### **Step 3: Preprocess Video Frames**

Frames were resized and normalized for AI model processing.

##### **Step 4: Apply YOLOv8 Detection**

YOLOv8 detected persons from real-time frames.

##### **Step 5: Perform Crowd Analysis**

The system counted detected persons to identify crowd conditions.

##### **Step 6: Detect Fallen Person**

Body posture analysis was performed for fall detection.

##### **Step 7: Detect SOS Gesture**

MediaPipe landmarks were analyzed to identify emergency gestures.

##### **Step 8: Generate Alerts and Display Output**

Emergency alerts and final detection results were displayed on the screen.

#### **4.12 Working Procedure**

The working procedure of the AI-Based Public Hazard Detection System starts when the webcam or surveillance camera captures live video frames continuously. The captured frames

are processed in real time using Computer Vision and Artificial Intelligence techniques for hazard detection and monitoring.

Initially, the captured video frames are preprocessed using OpenCV functions. The preprocessing stage includes frame resizing, image enhancement, and noise reduction to improve image quality and detection accuracy. After preprocessing, the frames are forwarded to the YOLOv8 Pose detection model for further analysis.

The YOLOv8 model detects humans present in the frame and generates bounding boxes around detected persons. The detected person information is then used for multiple hazard analysis operations such as crowd detection, fall detection, and SOS gesture recognition. The system continuously processes each frame to provide uninterrupted monitoring and surveillance.

Crowd detection is performed by counting the number of detected persons in each frame. If the number of detected individuals exceeds the predefined threshold value, the system identifies the condition as crowd formation and automatically generates a crowd warning alert. This helps in identifying overcrowded public environments in real time.

For fall detection, the system analyzes body posture and orientation using pose estimation techniques. The body dimensions and posture ratio are monitored continuously to identify fallen persons. Whenever the detected posture appears horizontal for a specific duration, the system classifies it as a fallen condition and generates an emergency warning message.

The SOS gesture detection module identifies emergency hand gestures using body landmark tracking and pose estimation. The system tracks body keypoints and checks whether both hands are raised in a specific emergency pattern. When the SOS gesture is detected successfully, the system automatically triggers an emergency alert message indicating that immediate assistance may be required.

The proposed system also includes an intelligent alert generation mechanism. Different warning banners and alert messages are displayed on the monitoring screen whenever hazardous situations are identified. Audio alerts are also generated to improve emergency awareness and response speed during critical situations.

The final processed output containing bounding boxes, hazard labels, person count, and warning messages is displayed continuously on the monitoring screen for real-time surveillance. The system operates automatically without manual intervention and provides continuous public safety monitoring support.

The overall working procedure demonstrates the effective integration of Artificial Intelligence, Deep Learning, and Computer Vision techniques for intelligent hazard detection and automated emergency monitoring applications.

## 4.13 Implementation

```
import cv2
import pygame
import os
from ultralytics import YOLO

# Configuration
CROWD_THRESHOLD = 5
FALLEN_RATIO = 1.0
CONFIDENCE_MIN = 0.5

# Alert Sound Files
CROWD_SOUND_FILE = "crowd.mp3"
FALLEN_SOUND_FILE = "alarm.wav"
HELP_SOUND_FILE = "help.mp3"

# YOLOv8 Pose Model Loading
model = YOLO("yolov8n-pose.pt")

# Webcam Connection
cap = cv2.VideoCapture(0)

# Audio Initialization
pygame.mixer.init()

# Audio Channels
ch_crowd = pygame.mixer.Channel(0)
ch_fall = pygame.mixer.Channel(1)
ch_sos = pygame.mixer.Channel(2)

while cap.isOpened():

    ret, frame = cap.read()

    if not ret:
        break

    # YOLO Detection
    results = model(frame, verbose=False)

    boxes = results[0].boxes
    keypoints = results[0].keypoints

    total_persons = 0
    fallen_count = 0
    sos_count = 0
```

```

if boxes is not None:

    total_persons = len(boxes)

    for i in range(len(boxes)):

        x1, y1, x2, y2 = map(int, boxes.xyxy[i])

        width = x2 - x1
        height = y2 - y1

        ratio = width / height if height > 0 else 0

        # Fall Detection
        is_fallen = ratio > FALLEN_RATIO

        if is_fallen:
            fallen_count += 1

        # SOS Detection
        is_sos = False

        if keypoints is not None:

            kpts = keypoints.data[i]

            ls_x, ls_y, ls_c = kpts[5]
            rs_x, rs_y, rs_c = kpts[6]
            lw_x, lw_y, lw_c = kpts[9]
            rw_x, rw_y, rw_c = kpts[10]

            if all(c > CONFIDENCE_MIN for c in [ls_c, rs_c, lw_c, rw_c]):

                hands_up = (lw_y < ls_y) and (rw_y < rs_y)

                shoulder_width = abs(ls_x - rs_x)

                wrist_distance = abs(lw_x - rw_x)

                if hands_up and wrist_distance < (shoulder_width * 0.8):

                    is_sos = True
                    sos_count += 1

        # Bounding Box Drawing
        if is_sos:

            color = (255, 0, 255)
            label = "SOS DETECTED"

        elif is_fallen:

            color = (0, 0, 255)
            label = "FALLEN PERSON"

        else:

            color = (0, 255, 0)
            label = "PERSON"

        cv2.rectangle(frame, (x1, y1), (x2, y2), color, 2)

        cv2.putText(
            frame,
            label,
            (x1, y1 - 10),
            cv2.FONT_HERSHEY_SIMPLEX,
            0.6,
            color,
            2
        )

```

```

# Crowd Detection
is_crowd = total_persons >= CROWD_THRESHOLD

if is_crowd:

    cv2.putText(
        frame,
        "CROWD DETECTED",
        (20, 40),
        cv2.FONT_HERSHEY_SIMPLEX,
        1,
        (0, 140, 255),
        2
    )

# Information Display
cv2.putText(
    frame,
    f"Persons: {total_persons}",
    (20, 80),
    cv2.FONT_HERSHEY_SIMPLEX,
    0.7,
    (255, 255, 255),
    2
)

cv2.putText(
    frame,
    f"Fallen: {fallen_count}",
    (20, 110),
    cv2.FONT_HERSHEY_SIMPLEX,
    0.7,
    (255, 255, 255),
    2
)

cv2.putText(
    frame,
    f"SOS: {sos_count}",
    (20, 140),
    cv2.FONT_HERSHEY_SIMPLEX,
    0.7,
    (255, 255, 255),
    2
)

# Output Window
cv2.imshow(
    "AI-Based Public Hazard Detection System",
    frame
)

# Exit Key
if cv2.waitKey(1) & 0xFF == ord('q'):
    break

# Release Resources
cap.release()
cv2.destroyAllWindows()
pygame.mixer.quit()

```

## CHAPTER 5

### RESULTS & TESTING

#### 5.1 Testing Environment

The system was tested under the following environment:

- Programming Language: Python 3.x
- Frameworks/Libraries: OpenCV, YOLOv8, MediaPipe, Pygame
- Input Device: Webcam (Live Video Feed)
- Operating System: Windows
- Hardware: Intel i5 processor, 8GB RAM

The system was executed in real-time conditions to evaluate performance and accuracy.

#### 5.2 Person Detection Results

The YOLOv8 model successfully detected persons in real-time video frames. Each detected person was highlighted using bounding boxes. The system performed efficiently under normal lighting conditions and maintained stable detection accuracy.

The results show that the system can continuously track multiple individuals in a frame without interruption.

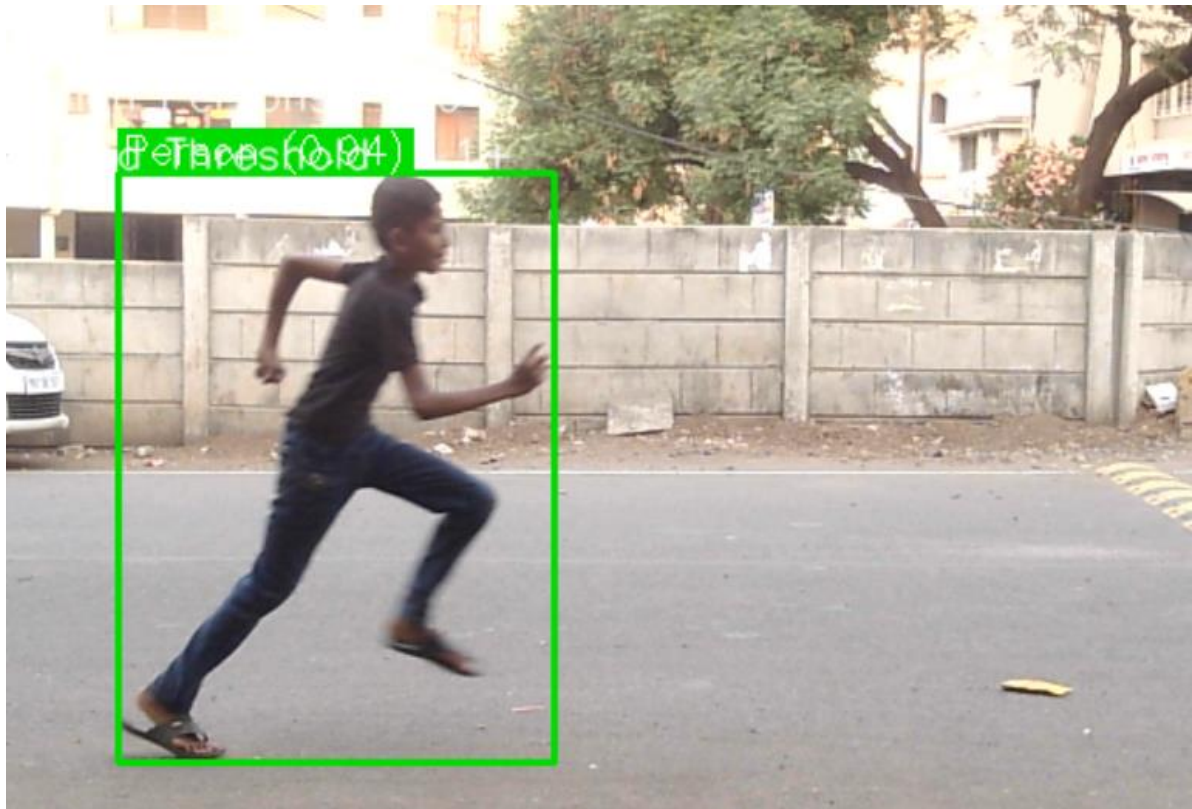


Fig 5.2.1: Real-Time Person Detection 1

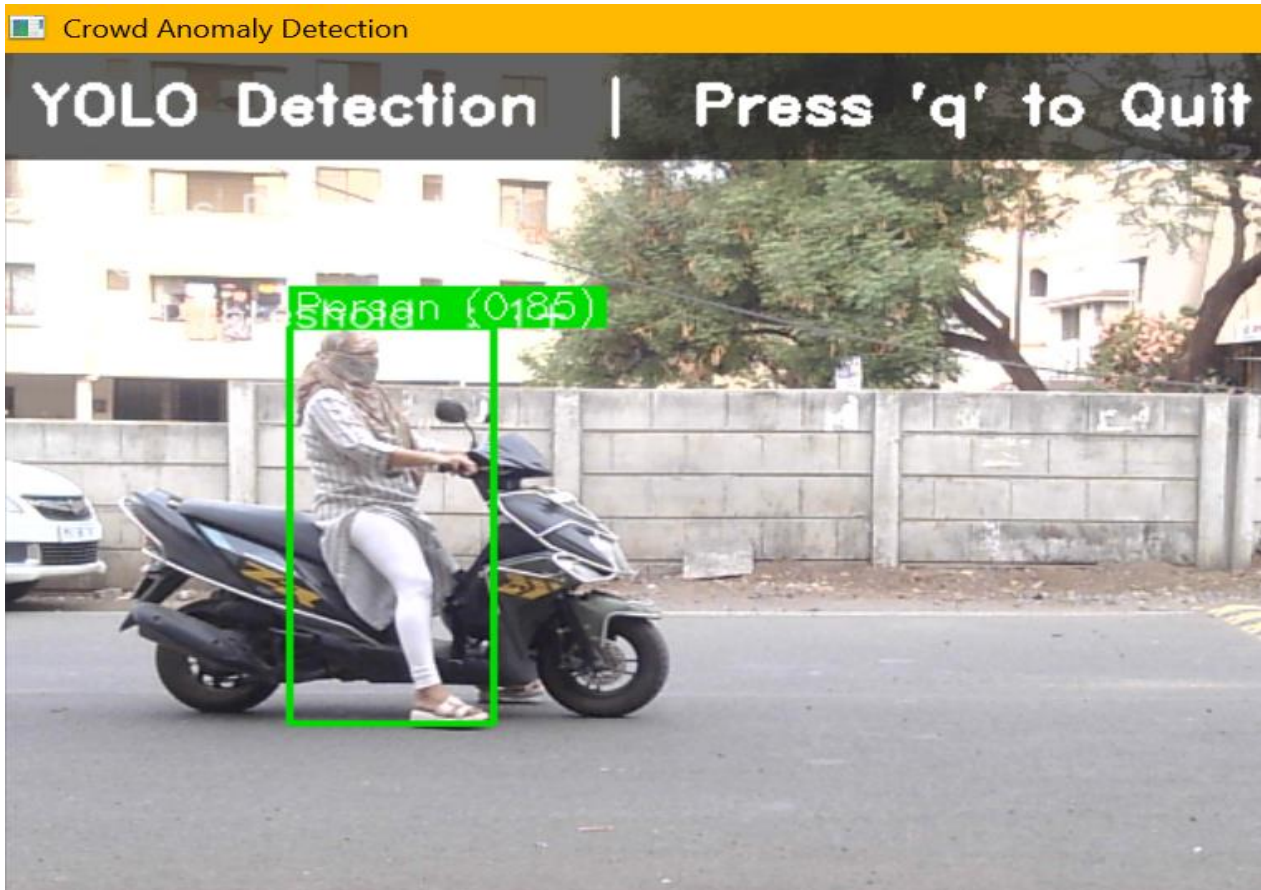


Fig 5.2.2: Real-Time Person Detection 2

### 5.3 Crowd Detection Results

Crowd detection was performed by counting the number of detected persons in each frame. When the number of persons exceeded the predefined threshold, the system generated a crowd alert.



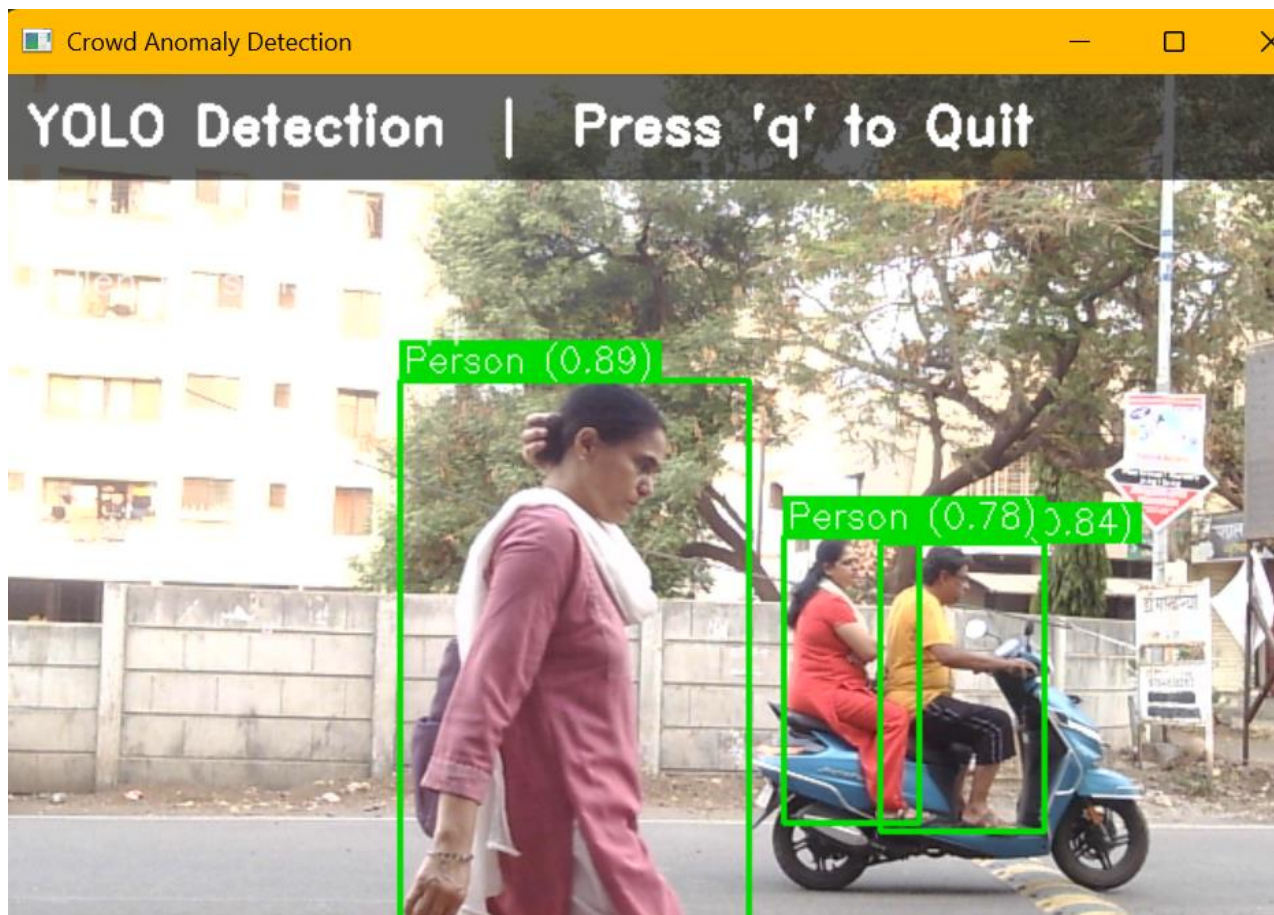


Fig 5.3.1: Real-Time Crowd Detection 1

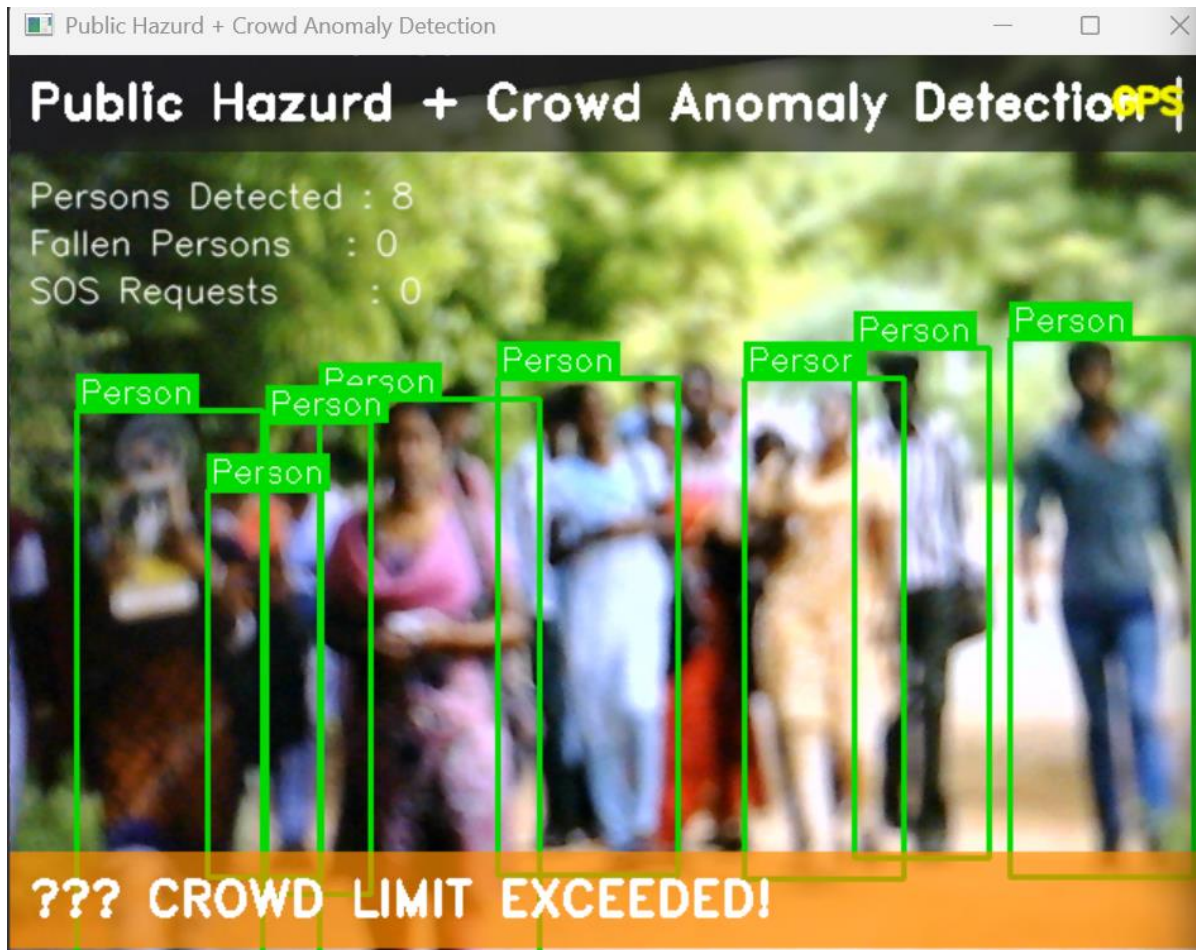


Fig 5.3.2: Real-Time Crowd Detection 2

The system successfully identified crowded environments and displayed warning messages on the screen.

#### 5.4 Fall Detection Results

The system was tested for fallen person detection using posture analysis. When a person appeared in a horizontal position for a certain duration, the system classified it as a fallen condition.

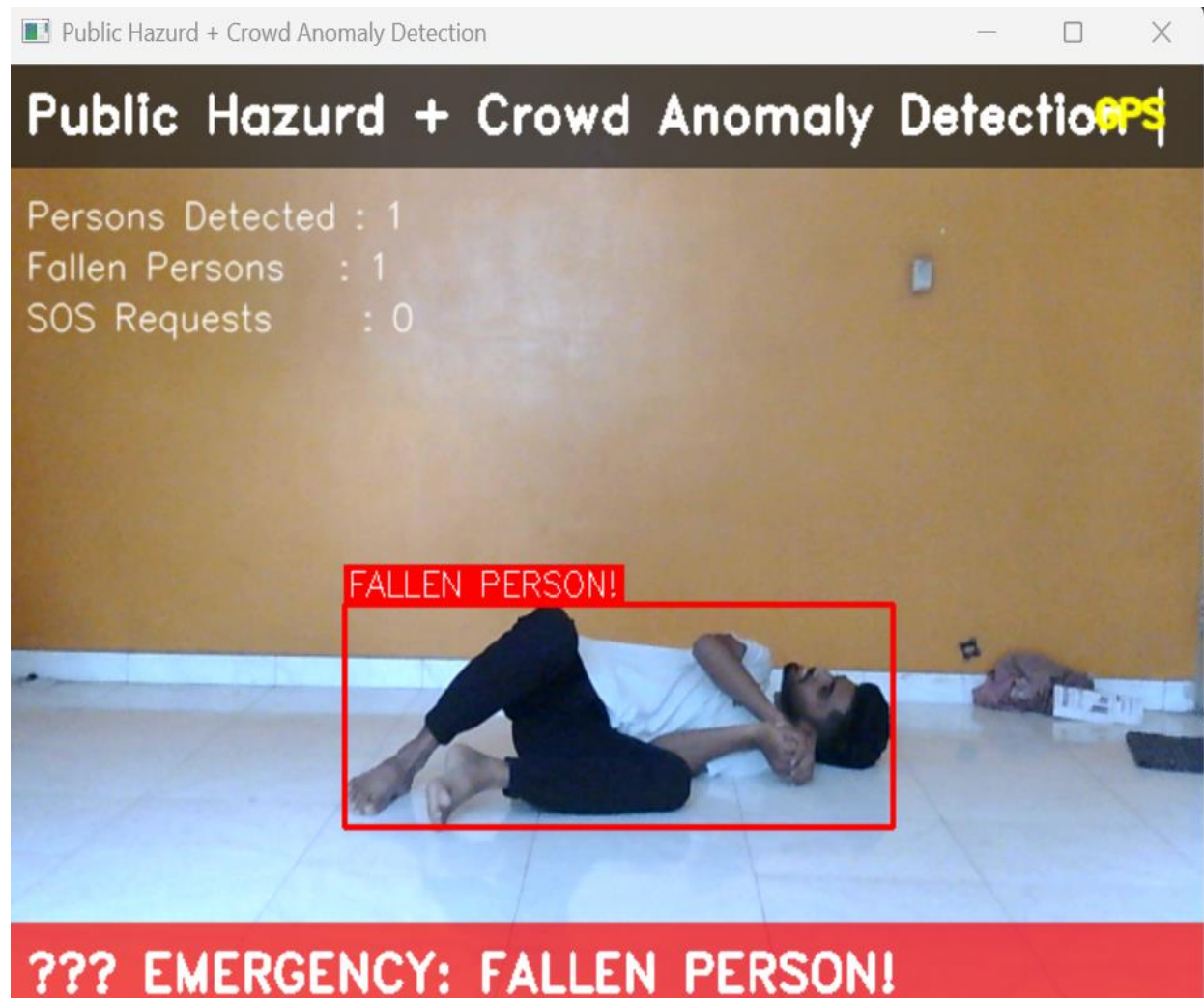


Fig 5.4.1: Fall Detection 1





Fig 5.4.2: Fall Detection 2

The results show that the system successfully detects fall scenarios and generates emergency alerts in real time

### 5.5 SOS Gesture Detection Results

The SOS detection module identifies emergency hand gestures using MediaPipe pose estimation. When a user raises both hands in a specific pattern, the system detects it as an SOS signal.

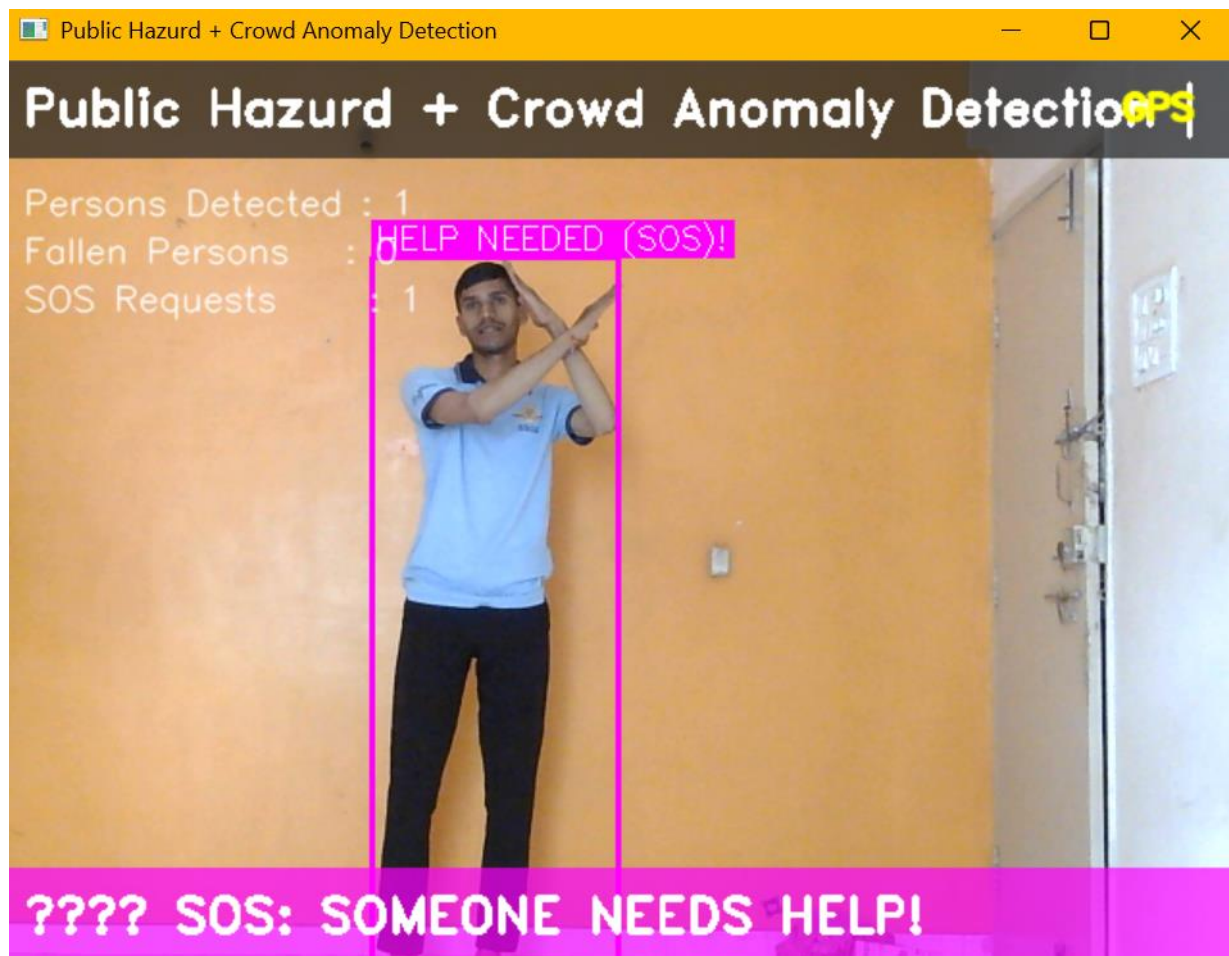


Fig 5.5.1: Real-Time SOS Gesture Detection 1



Fig 5.5.2: Real-Time SOS Gesture Detection 2

The system successfully identified SOS gestures and triggered emergency alerts during testing.

### 5.6 Alert System Results

The alert generation system plays an important role in the proposed AI-Based Public Hazard Detection System. The system continuously monitors live video streams and generates warning messages whenever hazardous conditions are detected. Real-time alerts help improve emergency response and public safety monitoring.

The system successfully generated alerts for crowd detection, fall detection, and SOS emergency gesture detection during testing.

### Crowd Detection Alert

The crowd alert system was tested under different crowd conditions. When the number of detected persons remained below the predefined threshold value, the system operated under normal monitoring mode.

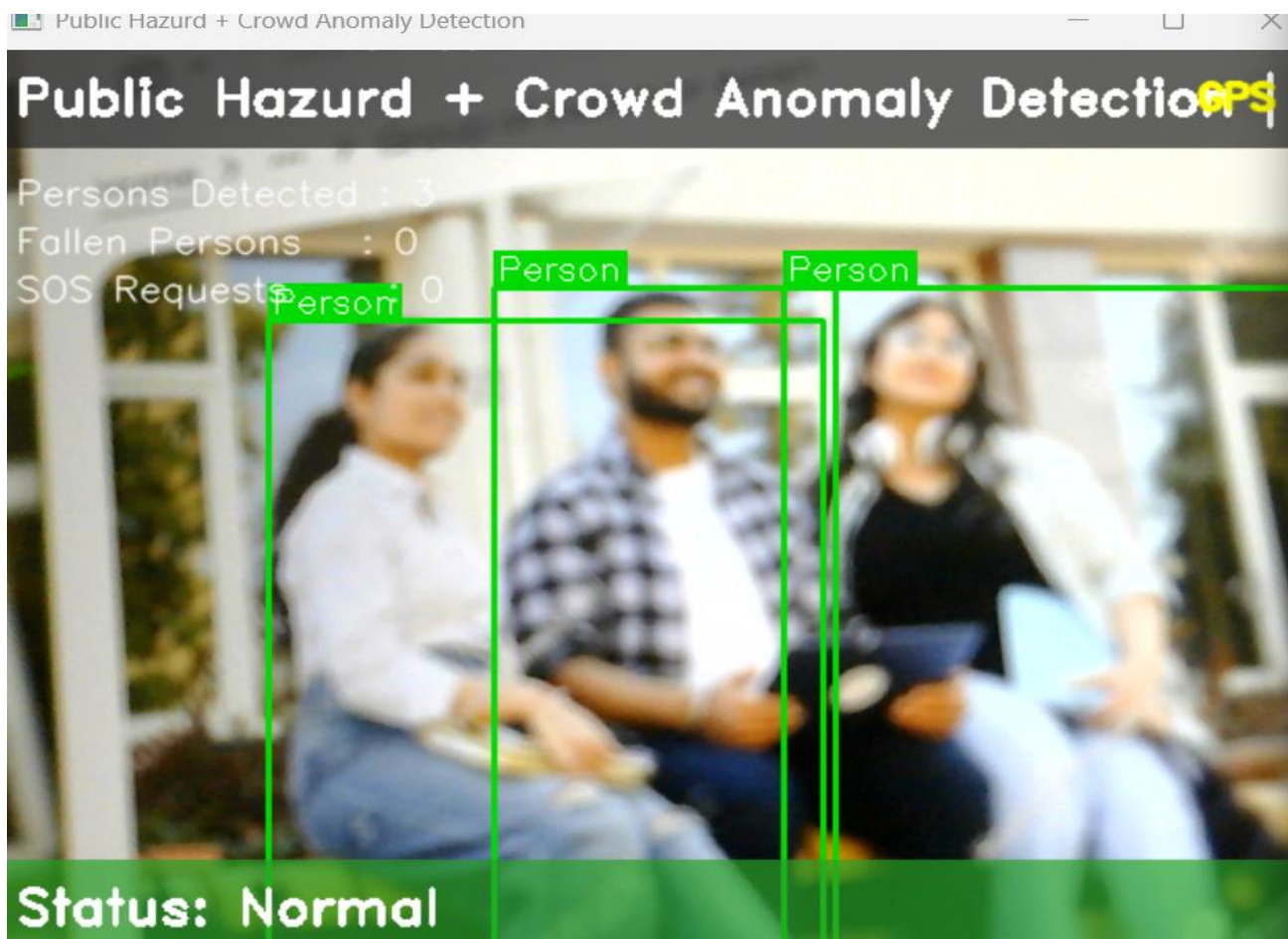


Fig 5.6.1: Crowd Detection Under Normal Conditions

Whenever the detected crowd count exceeded the threshold limit, the system automatically generated a crowd warning alert on the screen. The alert banner helped identify overcrowded situations in real time.



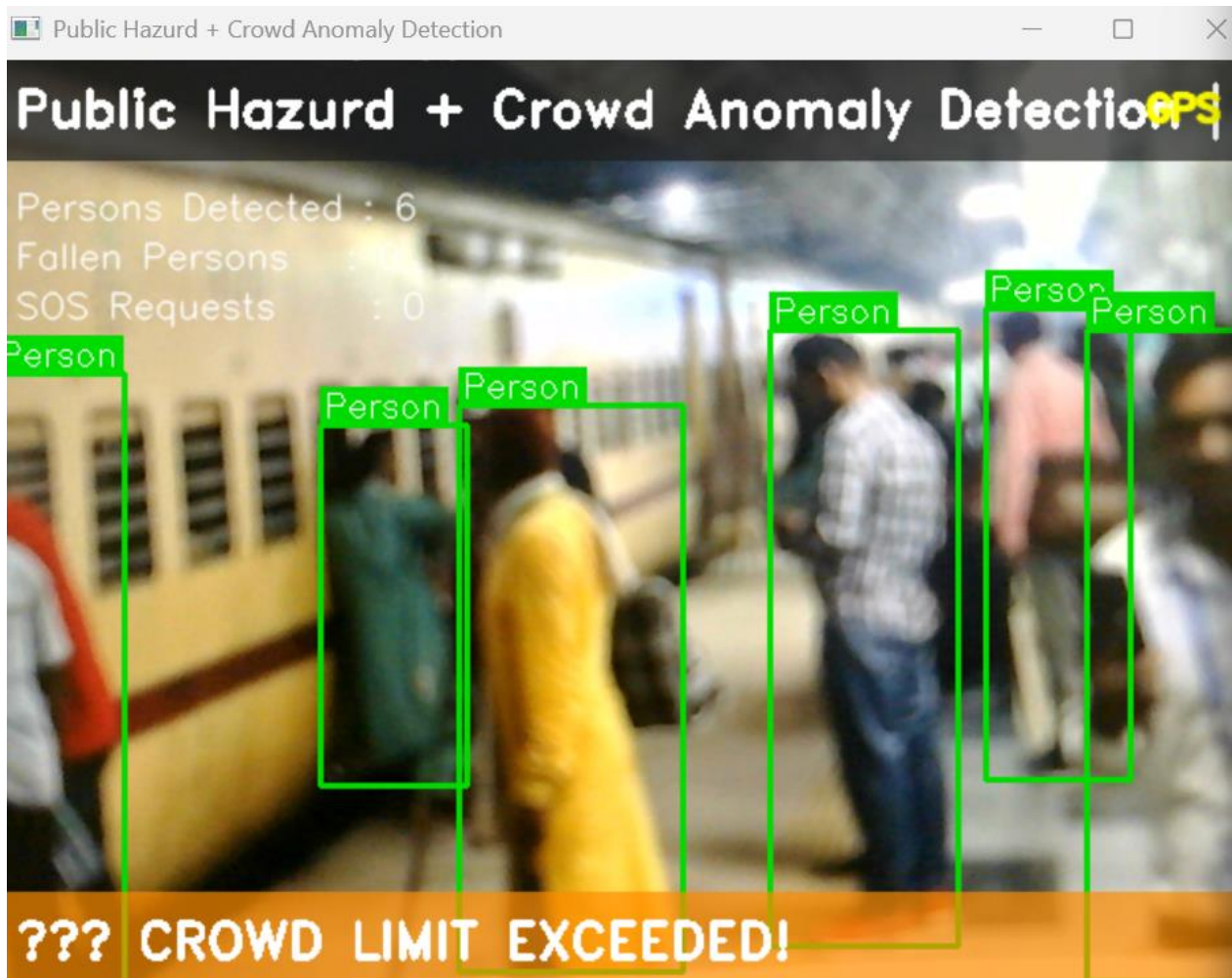


Fig 5.6.2: Crowd Limit Exceeded Alert

### Fall Detection Alert

The fall detection alert system successfully identified fallen persons using body posture analysis. Under normal conditions, the system detected standing individuals without generating any warning message.



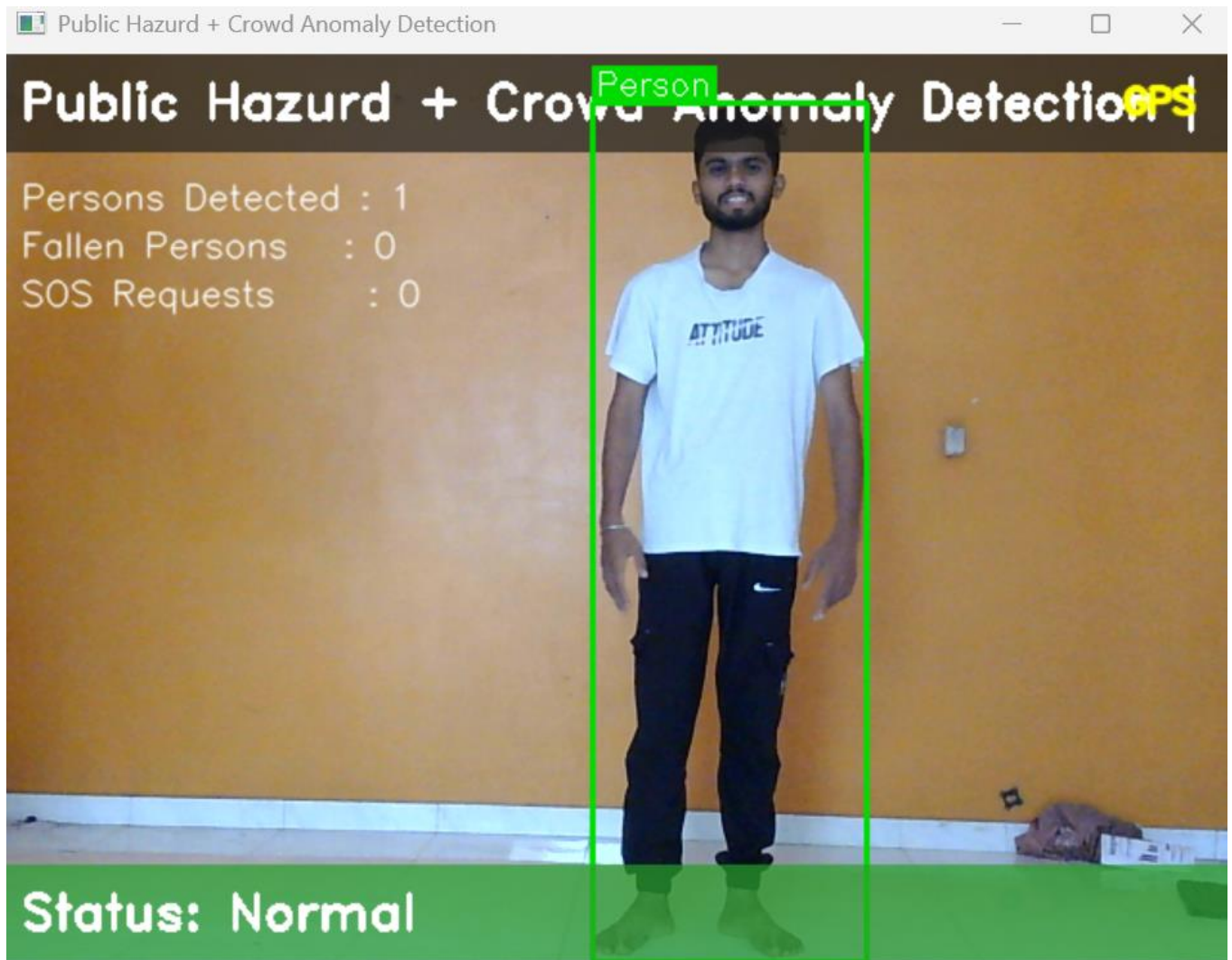


Fig 5.6.3: Normal Person Detection

When a person appeared in a fallen position for a specific duration, the system generated an emergency fall alert automatically. Warning banners and detection labels were displayed clearly on the monitoring screen.

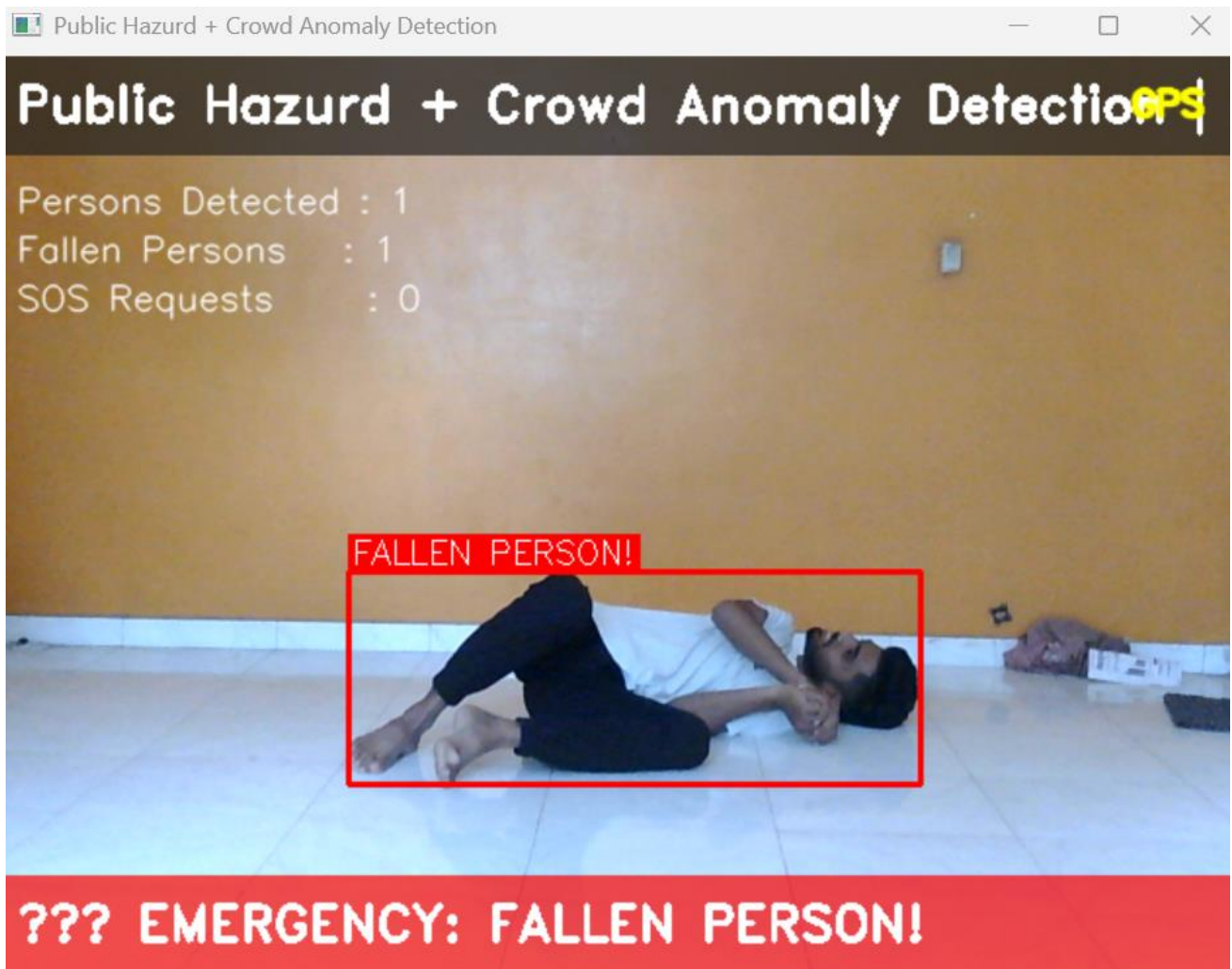


Fig 5.6.4: Fallen Person Alert

### **SOS Gesture Detection Alert**

The SOS detection system identified emergency gestures using body landmark analysis and pose estimation techniques. During normal conditions, the system continuously monitored human body movements without generating alerts.

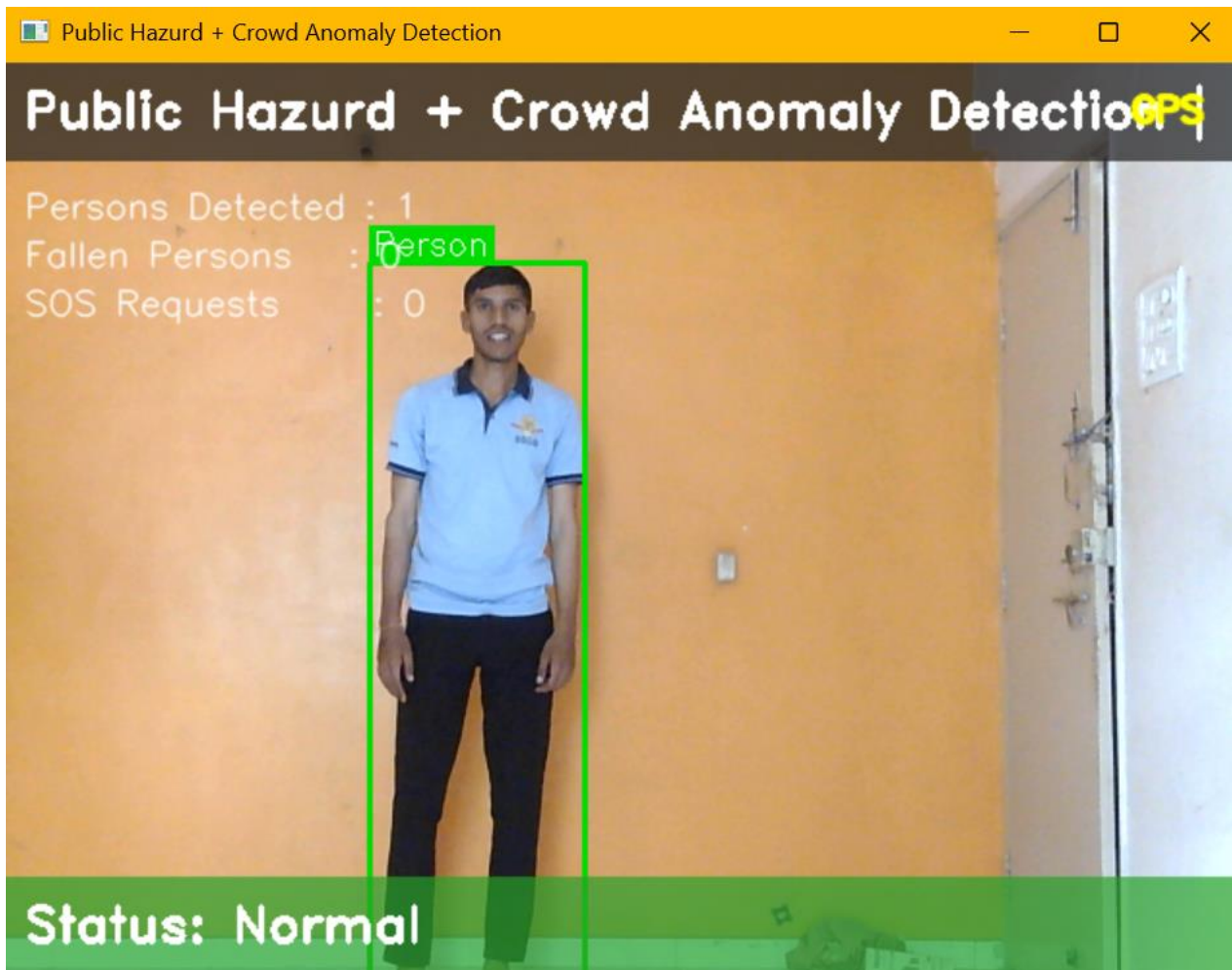


Fig 5.6.5: Normal Monitoring

When the SOS gesture was detected, the system immediately generated an emergency warning message indicating that assistance was required. The alert system successfully displayed SOS warning banners in real time.

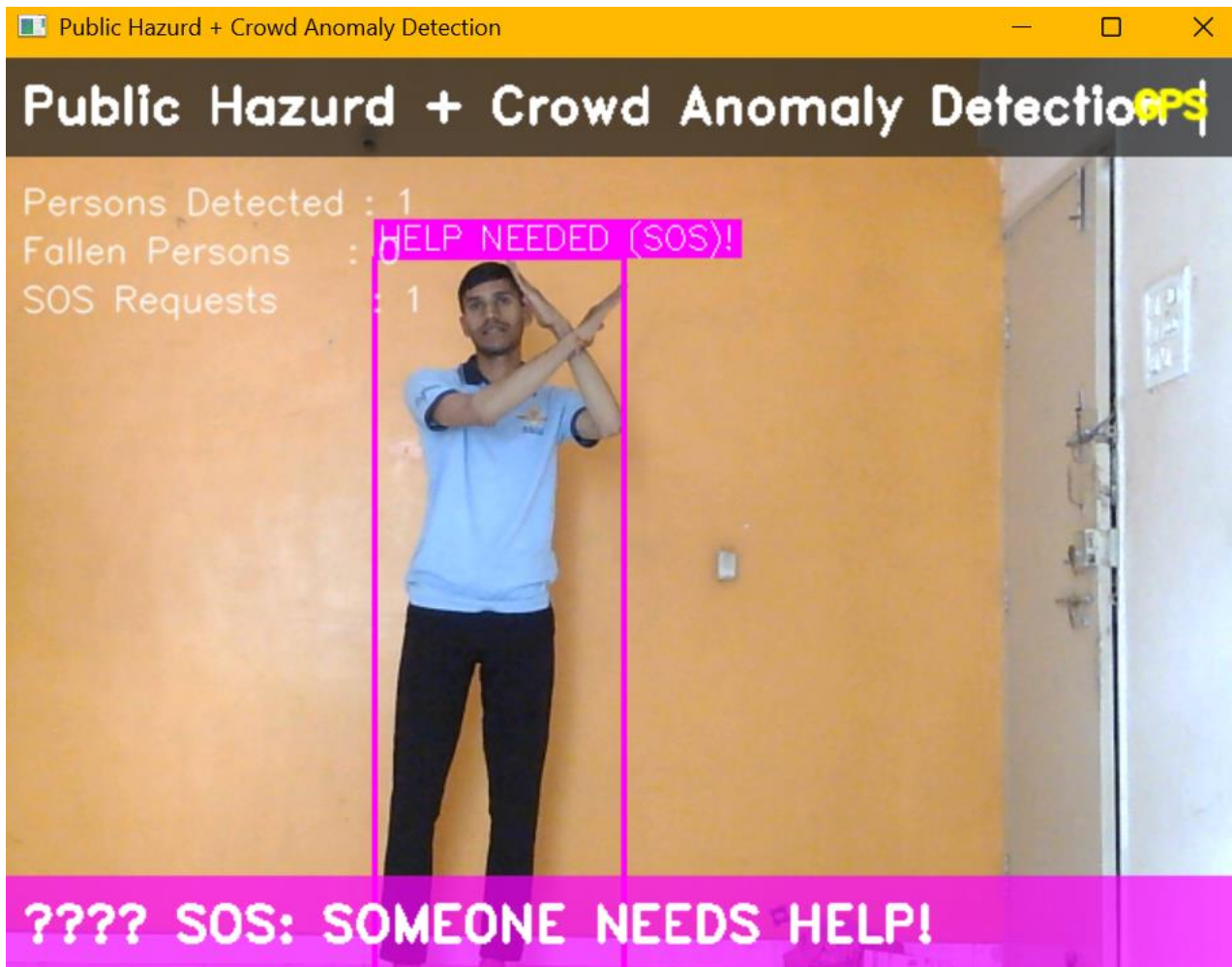


Fig 5.6.6: SOS Emergency Alert

### 5.7 Performance Analysis

The performance of the AI-Based Public Hazard Detection System was evaluated under real-time testing conditions using webcam video input. The system was tested for person detection, crowd detection, fall detection, and SOS gesture recognition.

The YOLOv8 model provided fast and accurate person detection with low processing delay. MediaPipe pose estimation successfully identified body posture and emergency gestures in real time. The system continuously processed live video frames and generated alerts efficiently.

The use of optimized processing techniques improved overall system performance and allowed continuous monitoring without major delays. The system maintained stable performance under normal lighting conditions and moderate crowd density. The alert generation system responded immediately whenever hazardous situations were detected. The proposed system also demonstrated stable real-time monitoring performance during continuous execution and live surveillance testing.

| Parameter            | Performance                  |
|----------------------|------------------------------|
| Person Detection     | High Accuracy                |
| Crowd Detection      | Real-Time Detection          |
| Fall Detection       | Fast Alert Response          |
| SOS Detection        | Accurate Gesture Recognition |
| Processing Speed     | Smooth Performance           |
| Alert Generation     | Immediate Response           |
| Real-Time Monitoring | Stable Execution             |

Table 5.7: Performance Evaluation

## 5.8 Test Case Evaluation

| Test Case ID | Feature          | Expected Output      | Actual Output       | Status |
|--------------|------------------|----------------------|---------------------|--------|
| TC-01        | Person Detection | Detect Person        | Person Detected     | Pass   |
| TC-02        | Crowd Detection  | Generate Alert       | Alert Generated     | Pass   |
| TC-03        | Fall Detection   | Detect Fallen Person | Person Detected     | Pass   |
| TC-04        | SOS Detection    | Detect SOS Gesture   | SOS Alert Generated | Pass   |

Table 5.8: Test Case Evaluation



## **CHAPTER 6**

### **CONCLUSION**

The AI-Based Public Hazard Detection System was successfully designed, developed, and implemented using Artificial Intelligence, Deep Learning, and Computer Vision technologies. The primary objective of the project was to create an intelligent real-time surveillance system capable of automatically detecting hazardous situations and improving public safety monitoring. The proposed system demonstrated effective performance in detecting multiple public hazard scenarios such as person detection, crowd detection, fall detection, and SOS emergency gesture recognition using live video streams.

The system was developed using Python programming language along with advanced Computer Vision libraries and AI frameworks such as OpenCV, YOLOv8, and MediaPipe. YOLOv8 was used for accurate real-time human detection and object localization, while MediaPipe pose estimation techniques were used for analyzing body posture and identifying emergency hand gestures. OpenCV functions were used for image preprocessing, frame handling, drawing bounding boxes, displaying warning messages, and performing real-time monitoring operations.

The proposed surveillance model continuously captures live video frames through a webcam or surveillance camera and processes them in real time. Each captured frame is analyzed using the YOLOv8 Pose detection model to identify humans present in the frame. After detecting persons, the system performs crowd analysis by counting the total number of detected individuals. If the detected crowd count exceeds the predefined threshold value, the system automatically generates crowd warning alerts to indicate overcrowded public conditions.

The fall detection module was implemented using posture ratio analysis and body orientation monitoring techniques. The system continuously analyzes body dimensions and posture positions to determine whether a person has fallen. Whenever the detected posture appears horizontal for a specific period of time, the system identifies the condition as a fallen person scenario and generates emergency alerts automatically. This functionality can help improve accident monitoring and emergency response operations in surveillance environments.

The SOS gesture recognition module was implemented using body landmark tracking and pose estimation methods. The system tracks body keypoints and detects emergency hand gestures by analyzing hand positions relative to shoulder positions. When a user performs the SOS gesture pattern, the system immediately identifies the gesture and generates emergency

warning messages indicating that assistance may be required. This feature improves emergency communication capabilities and supports intelligent surveillance monitoring in critical situations.

The alert generation system also played an important role in the successful implementation of the project. Whenever hazardous conditions such as crowd formation, fallen persons, or SOS gestures were detected, the system generated real-time warning banners and emergency notifications on the monitoring screen. Audio alerts were also generated during emergency situations to improve awareness and response speed. The alert mechanism improved the efficiency of the overall monitoring process and reduced the dependency on continuous human observation.

The implementation results showed that the system maintained stable performance during continuous real-time monitoring and testing conditions. The proposed model successfully processed live video streams and generated accurate detection results with minimal processing delay. The integration of Artificial Intelligence techniques improved system automation, detection efficiency, and monitoring accuracy. The system also demonstrated effective performance under normal lighting conditions and moderate crowd environments.

The project successfully reduced manual monitoring effort by automating hazard detection and emergency identification processes. Traditional surveillance systems generally require continuous human supervision, which may lead to delayed responses and monitoring inefficiencies. The proposed AI-based system minimized these limitations by providing automated real-time monitoring and intelligent hazard recognition capabilities. The developed model can support public safety operations in crowded environments, institutions, transportation areas, surveillance systems, and other monitoring applications.

The development process of the project also helped demonstrate the practical application of Deep Learning and Computer Vision techniques in real-world surveillance systems. The implementation of YOLOv8, pose estimation, real-time frame processing, and intelligent alert generation highlighted the importance of AI-based technologies in modern smart monitoring systems. The project provided practical knowledge related to Artificial Intelligence model integration, image processing, surveillance automation, and emergency response systems.

The proposed AI-Based Public Hazard Detection System achieved all major objectives defined during the project planning phase. The system successfully implemented real-time person detection, crowd monitoring, fall detection, SOS gesture recognition, and automated alert generation using Artificial Intelligence techniques. The project demonstrated that AI-powered

surveillance systems can significantly improve monitoring efficiency, emergency response speed, and intelligent public safety operations.

Overall, the AI-Based Public Hazard Detection System provides an efficient, intelligent, and practical solution for automated surveillance and hazard monitoring applications. The integration of Artificial Intelligence, Deep Learning, and Computer Vision technologies improved the capability of the system to monitor public environments continuously and identify emergency situations automatically. The project demonstrates the growing importance of AI-driven surveillance systems in developing modern smart safety and security applications for real-time public hazard monitoring and emergency management.



## **CHAPTER 7**

### **FUTURE SCOPE**

The AI-Based Public Hazard Detection System provides an intelligent and automated approach for real-time surveillance, hazard monitoring, and emergency detection. The proposed system successfully demonstrated the capability of detecting persons, identifying crowd conditions, recognizing fallen persons, and detecting SOS emergency gestures using Artificial Intelligence and Computer Vision technologies. Although the developed model achieved effective real-time monitoring performance, several advanced improvements and additional functionalities can be integrated in the future to enhance system accuracy, scalability, automation, and real-world implementation capabilities.

One of the major future enhancements of the proposed system is the integration of cloud-based surveillance and centralized monitoring platforms. Cloud integration will allow surveillance data and alert information to be stored and accessed remotely through internet-based systems. This feature can support centralized monitoring of multiple surveillance locations simultaneously and improve large-scale public safety management. Cloud connectivity can also enable remote access to hazard reports, monitoring dashboards, and real-time alert notifications from different locations.

Another important future improvement is the development of mobile application support for emergency monitoring and remote surveillance operations. Real-time notifications related to crowd alerts, fallen person detection, and SOS emergency gestures can be sent directly to smartphones or mobile devices of security personnel and authorized authorities. Mobile application integration can improve emergency response time and increase the effectiveness of public hazard management systems.

The current implementation of the proposed system mainly focuses on a single surveillance feed. In the future, the system can be expanded to support multi-camera surveillance and large-scale monitoring environments. Multiple surveillance cameras can be integrated together to monitor different locations simultaneously. Such improvements can make the system suitable for railway stations, airports, shopping malls, educational institutions, public transport systems, hospitals, and smart city surveillance applications.

Future versions of the system can also include advanced Deep Learning models trained using larger and more diverse datasets. The current detection model performs efficiently under normal testing conditions, but additional training data collected under different lighting

conditions, camera angles, weather conditions, and complex crowd environments can further improve detection accuracy and reduce false detections. More advanced AI architectures can also improve real-time hazard recognition efficiency and surveillance reliability.

Another possible enhancement is the implementation of suspicious activity detection and abnormal behavior analysis. Future systems can automatically identify violence, fights, unattended objects, running behavior, restricted area access, or unusual public movement patterns. Such intelligent anomaly detection systems can significantly improve automated public safety operations and advanced surveillance management capabilities.

The proposed system can also be integrated with Internet of Things (IoT) technologies for developing smart emergency response systems. Smart alarms, emergency lights, warning systems, and automated communication devices can be connected with the surveillance model to provide faster and more effective emergency response operations. IoT integration can improve system automation and create intelligent interconnected surveillance networks for modern smart environments.

Future improvements may also include advanced night vision and low-light hazard detection techniques. The current system performs efficiently under normal lighting conditions, but surveillance accuracy may decrease in dark or poor visibility environments. Integration of infrared cameras, thermal imaging systems, or advanced low-light AI models can improve night-time surveillance performance and emergency monitoring capabilities.

The implementation of edge computing and embedded AI devices is another important future scope of the project. The proposed model can be deployed on compact embedded systems such as Raspberry Pi, NVIDIA Jetson Nano, or other AI-enabled edge devices. Edge deployment can reduce system cost, improve portability, decrease processing delay, and support real-time hazard monitoring without requiring high-end computing systems.

Another future enhancement involves the integration of facial recognition and identity verification systems. Face recognition technologies can help identify missing persons, suspicious individuals, or unauthorized persons in restricted areas. Combining face recognition with hazard detection systems can significantly improve intelligent surveillance and public safety monitoring operations.

The system can also be improved by adding automated reporting and data analytics features. Hazard event logs, alert history, person count statistics, surveillance summaries, and graphical monitoring reports can be generated automatically for administrative and security analysis purposes. Such features can improve surveillance management, decision-making processes, and long-term safety planning operations.

Future systems can further support real-time communication with emergency departments, security teams, and public safety authorities. Whenever hazardous situations are detected, automated alerts and emergency notifications can be transmitted directly to response teams for immediate action. Such integration can improve emergency handling efficiency and reduce response time during critical situations.

The proposed AI-Based Public Hazard Detection System also has strong potential for integration with smart city infrastructure and intelligent urban surveillance systems. Artificial Intelligence-driven monitoring systems can help improve traffic monitoring, public crowd management, disaster response operations, and intelligent security systems for modern urban environments. The continuous advancement of AI technologies will further improve the scalability, automation, and intelligence of surveillance systems in the future.

The project also opens opportunities for future research in the fields of Deep Learning, Computer Vision, intelligent surveillance systems, and automated emergency response technologies. Researchers and developers can further improve detection algorithms, optimize processing speed, reduce computational complexity, and develop more advanced hazard recognition techniques using modern AI architectures.

Overall, the future scope of the AI-Based Public Hazard Detection System is extensive and highly promising. The integration of Artificial Intelligence, Deep Learning, Computer Vision, IoT, and smart surveillance technologies can help develop highly advanced and intelligent public safety systems capable of providing continuous automated monitoring, accurate hazard detection, faster emergency response, and improved surveillance efficiency. The proposed system represents an important step toward the development of intelligent smart surveillance solutions for modern public safety and security applications.

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